

WEYL–CHAMBER GEOMETRY, POINCARÉ INEQUALITY MEETS SCHATTEN CLASS PROPERTIES OF DUNKL–RIESZ TRANSFORM COMMUTATORS

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ABSTRACT.

CONTENTS

1. Introduction	1
2. Preliminaries	2
2.1. Harmonic analysis in the Dunkl setting	2
2.2. Weyl chamber as simplicial cone	3
2.3. System of dyadic cubes	3
2.4. Nearly weakly orthonormal sequences	3
2.5. Principal symbol and crossed products	4
3. Chamber is a doubling metric measure space	5
4. Rochberg-Semmes based estimate	7
5. Proof of special Cwikel estimate in Dunkl setting	10
6. Proof of Theorem 1.1 (i)	12
7. Proof of Theorem 1.1 (ii)	14
8. Proof of smoothness	19
9. Approximation theorem	23
9.1. First approximation theorem	24
9.2. Second approximation theorem	25
9.3. Proof of Theorem 9.1	27
10. Spectral asymptotic formula	30
Appendix A. Zhijie lemma	32
References	33

1. INTRODUCTION

Theorem 1.1. *Let R be a normalised root system in $\mathbb{R}^N$, $N \geq 2$, and let G be the corresponding finite reflection group. Let $\kappa : R \rightarrow \mathbb{R}_+$ be G -invariant multiplicity function. Let $R_{\text{Dunkl},j}$ be the j -th Dunkl-Riesz transform corresponding to R and k . Let w be the Dunkl weight corresponding to R and k . Let $f \in L_1^{\text{loc}}(\mathbb{R}^N)$.*

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- (i) If $f \in \dot{W}^{1,N}(\mathbb{R}^N)$, then $[R_{\text{Dunkl},j}, M_f] \in \mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))$. Furthermore,

$$\|[R_{\text{Dunkl},j}, M_f]\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))} \leq c_\kappa \|f\|_{\dot{W}^{1,N}(\mathbb{R}^N)};$$

- (ii) If $[R_{\text{Dunkl},j}, M_f] \in \mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))$, then $f \in \dot{W}^{1,N}(\mathbb{R}^N)$. Furthermore,

$$\|[R_{\text{Dunkl},j}, M_f]\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))} \geq c_\kappa^{-1} \|f\|_{\dot{W}^{1,N}(\mathbb{R}^N)};$$

2. PRELIMINARIES

2.1. Harmonic analysis in the Dunkl setting. Let $\mathbb{R}^N$ be the Euclidean space equipped with the usual scalar product $\langle \cdot, \cdot \rangle$ and the induced norm $\|\cdot\|$. For a nonzero vector $v \in \mathbb{R}^N$, the reflection σ_v with respect to the hyperplane orthogonal to a normalised vector v is defined by

$$\sigma_v x := x - 2\langle x, v \rangle v.$$

A finite set $R \subset \mathbb{R}^N \setminus \{0\}$ is called a root system if $\sigma_v(R) = R$ for every $v \in R$. A root system R is called a normalised reduced root system if $\|v\|^2 = 1$ for every $v \in R$. The finite group G generated by the reflections $\{\sigma_v\}_{v \in R}$ is called the Weyl group (reflection group) of the root system R .

Let $\kappa : R \rightarrow [0, \infty)$ be a multiplicity function, which is a G -invariant function fixed throughout the paper.

Define $d(x, y) = \min_{g \in G} |x - gy|$. Recall from [7, Chapter VII, proof of Theorem 2.12] that $d(x, y) = |x - y|$ iff x and y belong to the same chamber.

Define the Dunkl weight on $\mathbb{R}^N$ by

$$w(x) = \prod_{v \in R} |\langle x, v \rangle|^{\kappa(v)}, \quad x \in \mathbb{R}^N.$$

Dunkl measure $w(x)dx$ is also denoted by w .

Let $B(x, r) := \{y \in \mathbb{R}^N : \|x - y\| < r\}$ be the Euclidean ball centred at x with radius $r > 0$. Let $\mathbf{N} := N + \sum_{v \in R} \kappa(v)$. Note that

$$w(B(tx, tr)) = t^{\mathbf{N}} w(B(x, r)), \quad \text{for } x \in \mathbb{R}^N, \quad t, r > 0.$$

Furthermore,

$$(1) \quad w(B(x, r)) \sim r^{\mathbf{N}} \prod_{v \in R} (|\langle x, v \rangle| + r)^{\kappa(v)},$$

where the implicit constant only depends on N, R and κ . It follows from (1) that there exists a constant $C \geq 1$ such that for every $x \in \mathbb{R}^N$,

$$(2) \quad C^{-1} \left(\frac{r_2}{r_1} \right)^{\mathbf{N}} \leq \frac{w(B(x, r_2))}{w(B(x, r_1))} \leq C \left(\frac{r_2}{r_1} \right)^{\mathbf{N}}, \quad \text{for } 0 < r_1 < r_2,$$

so the numbers $\mathbf{N}$ and N are called the upper and lower homogeneous dimensions, respectively.

The Dunkl operators T_ξ are defined by

$$T_\xi f(x) := \partial_\xi f(x) + \sum_{v \in R} \frac{\kappa(v)}{2} \langle v, \xi \rangle \frac{f(x) - f(\sigma_v(x))}{\langle v, x \rangle},$$

where $\partial_\xi f$ denotes the directional derivative of f in the direction ξ . Furthermore, the Dunkl Laplacian is defined by

$$\Delta_{\text{Dunkl}} = - \sum_{j=1}^N T_{e_j}^2.$$

In this article, we consider the Dunkl-Riesz transforms defined by $R_{\text{Dunkl},j} = T_{e_j} \Delta_{\text{Dunkl}}^{-\frac{1}{2}}$, $j = 1, 2, \dots, N$.

Operators $U_g : f \rightarrow f \circ g^{-1}$, $g \in G$, are unitary both in $L_2(\mathbb{R}^N)$ and in $L_2(\mathbb{R}^N, w(x)dx)$. Obviously, $g \rightarrow U_g$ is a unitary representation of the group G . To lighten the notations, we write U_v instead of U_{σ_v} for $v \in R$.

2.2. Weyl chamber as simplicial cone.

Definition 2.1. A connected component of the set

$$\{x \in \mathbb{R}^N : \langle x, v \rangle \neq 0 \text{ for all } v \in R\}$$

is called an open Weyl chamber. Its closure is called a Weyl chamber.

The following assertion can be found in e.g. Theorem 2 on p.166 in [1].

Theorem 2.2. Let Γ be a chamber. There exists a linear isomorphism $\Phi : \mathbb{R}^N \rightarrow \mathbb{R}^N$ such that $\Phi(\Gamma) = R_+^{N_1} \times \mathbb{R}^{N_2}$ (with $N_1 + N_2 = N$).

2.3. System of dyadic cubes.

Notation 2.3. Let

$$\mathbb{D} = \bigcup_{k \in \mathbb{Z}} \mathbb{D}_k$$

be the standard system of dyadic cubes over $\mathbb{R}^N$. Here each dyadic cube in $\mathbb{D}_k$ is of the shape

$$\frac{m}{2^k} + [0, 2^{-k})^N, \quad m \in \mathbb{Z}^N.$$

For every dyadic cube $Q \in \mathbb{D}$, we denote by c_Q and $\ell(Q)$ its center and side-length, respectively.

2.4. Nearly weakly orthonormal sequences. Rochberg and Semmes [12] introduced the notion of *nearly weakly orthonormal (NWO)* sequences of functions. This notion has since proved highly effective in characterizing Schatten class membership of singular integral commutators in a variety of contexts (see [9, 11, 3, 4, 10, 12, 14, 13]). The properties of NWO sequences were originally investigated in the Euclidean setting (equipped with Lebesgue measure), but the extension to spaces of homogeneous type is straightforward once a system of dyadic cubes is available in this general framework (see [9, Section 13] for a complete proof).

Definition 2.4 and Lemma 2.5 as stated below are my understanding of Hytonen's Corollary 8.2 and Lemma 8.7. Please check whether this understanding is the correct one!

Definition 2.4. Let $\mathbf{w} \in A_2(\mathbb{R}^N)$. Let $c > 0$ and $2 < r \leq \infty$. A sequence of functions $\{e_Q\}_{Q \in \mathbb{D}}$ is called (r, c) -NWO if

- (i) $\text{supp}(e_Q) \subseteq cQ$ for every $Q \in \mathbb{D}$;
- (ii) $\|e_Q\|_{L_r(\mathbb{R}^N, \mathbf{w}(x)dx)} \leq \mathbf{w}(Q)^{\frac{1}{r} - \frac{1}{2}}$ for every $Q \in \mathbb{D}$;

Lemma 2.5. *Let $\mathbf{w} \in A_2(\mathbb{R}^N)$. Let $0 < p < \infty$. Suppose that A is an integral operator on $L_2(\mathbb{R}^N, \mathbf{w}(x)dx)$ with integral kernel*

$$(x, y) \rightarrow \sum_{Q \in \mathbb{D}} \lambda_Q e_Q(x) f_Q(y), \quad x, y \in \mathbb{R}^N,$$

where $\{e_Q\}_{Q \in \mathbb{D}}$ and $\{f_Q\}_{Q \in \mathbb{D}}$ are (r_1, c_1) -NWO and (r_2, c_2) -NWO sequences, respectively. We have

$$\begin{aligned} \|A\|_{\mathcal{L}_{p,\infty}(L_2(\mathbb{R}^N, \mathbf{w}(x)dx))} &\leq C_{w,r_1,c_1,r_2,c_2,p} \|\{\lambda_Q\}_{Q \in \mathbb{D}}\|_{p,\infty}, \\ \text{dist}_{\mathcal{L}_{p,\infty}(L_2(\Gamma, \mathbf{w}(x)dx))} \left(A, (\mathcal{L}_{p,\infty})_0(L_2(\mathbb{R}^N, \mathbf{w}(x)dx)) \right) &\leq \\ &\leq C_{\mathbf{w},r_1,c_1,r_2,c_2,p} \text{dist}_{l_{p,\infty}(\mathbb{D})} \left(\{\lambda_Q\}_{Q \in \mathbb{D}}, (l_{p,\infty})_0(\mathbb{D}) \right). \end{aligned}$$

Proof. The assertion follows from Corollary 8.2 and Lemma 8.7 in [9]. $\square$

Lemma 2.6. *Let $w \in A_2(\mathbb{R}^N)$. Let $1 < p < \infty$. Suppose that A is an integral operator on $L_2(\mathbb{R}^N, \mathbf{w}(x)dx)$ with integral kernel*

$$(x, y) \rightarrow \sum_{Q \in \mathbb{D}} \lambda_Q e_Q(x) f_Q(y), \quad x, y \in \mathbb{R}^N,$$

where $\{e_Q\}_{Q \in \mathbb{D}}$ and $\{f_Q\}_{Q \in \mathbb{D}}$ are (r_1, c_1) -NWO and (r_2, c_2) -NWO sequences, respectively. We have

$$\left\| \left\{ \langle A e_Q, f_Q \rangle_{L_2(\mathbb{R}^N, \mathbf{w}(x)dx)} \right\}_{Q \in \mathbb{D}} \right\|_{l_{p,q}(\mathbb{D})} \leq c_{\mathbf{w},r_1,c_1,r_2,c_2,p,q} \|A\|_{\mathcal{L}_{p,q}(L_2(\mathbb{R}^N, \mathbf{w}(x)dx))}.$$

Proof. The assertion follows from Corollary 8.2 and Lemma 8.7 in [9]. $\square$

2.5. Principal symbol and crossed products. Let Π be the C^* -subalgebra in $B(L_2(\mathbb{R}^N))$ generated by M_f , $f \in (C_0 + \mathbb{C})(\mathbb{R}^N)$ and $h(\frac{\nabla}{\sqrt{\delta}})$, $h \in C(\mathbb{S}^{N-1})$. Note that every compact operator is contained in Π . There exists a surjective $*$ -homomorphism

$$\text{sym} : \Pi \rightarrow (C_0 + \mathbb{C})(\mathbb{R}^N) \otimes_{\min} C(\mathbb{S}^{N-1})$$

such that $\ker(\text{sym})$ consists exactly of all compact operators.

We have a natural action α of G on $B(L_2(\mathbb{R}^N))$ and, therefore, on Π — the C^* -algebra of operators admitting principal symbol. Namely,

$$\alpha(g) : T \rightarrow U_g T U_g^{-1}, \quad T \in B(L_2(\mathbb{R}^N)), \quad g \in G.$$

Define the action β of G on $(C_0 + \mathbb{C})(\mathbb{R}^N) \otimes_{\min} C(\mathbb{S}^{N-1})$ by setting

$$(\beta(g))(f \otimes h) = (f \circ g^{-1}) \otimes (h \circ g^{-1}).$$

The same action extends to von Neumann algebra $L_\infty(\mathbb{R}^N) \bar{\otimes} L_\infty(\mathbb{S}^{N-1})$. We have an equivariance relation

$$\text{sym}(\alpha(g)(T)) = (\beta(g))(\text{sym}(T)).$$

Clearly, sym is surjective, $\ker(\text{sym})$ is α -invariant. Hence, there exists a natural $*$ -homomorphism

$$\pi \rtimes G : \Pi \rtimes_\alpha G \rightarrow ((C_0 + \mathbb{C})(\mathbb{R}^N) \otimes_{\min} C(\mathbb{S}^{N-1})) \rtimes_\beta G.$$

We have an inclusion

$$((C_0 + \mathbb{C})(\mathbb{R}^N) \otimes_{\min} C(\mathbb{S}^{N-1})) \rtimes_\beta G \subset (L_\infty(\mathbb{R}^N) \bar{\otimes} L_\infty(\mathbb{S}^{N-1})) \rtimes_\beta G.$$

Hence,

$$\pi \rtimes G : \Pi \rtimes_{\alpha} G \rightarrow (L_{\infty}(\mathbb{R}^N) \bar{\otimes} L_{\infty}(\mathbb{S}^{N-1})) \rtimes_{\beta} G.$$

Next, the standard trace $\int_{\mathbb{R}^N} \otimes \int_{\mathbb{S}^{N-1}}$ on $L_{\infty}(\mathbb{R}^N) \bar{\otimes} L_{\infty}(\mathbb{S}^{N-1})$ is β -invariant. Hence, there exists a natural trace

$$\left(\int_{\mathbb{R}^N} \otimes \int_{\mathbb{S}^{N-1}} \right) \rtimes G$$

on $(L_{\infty}(\mathbb{R}^N) \bar{\otimes} L_{\infty}(\mathbb{S}^{N-1})) \rtimes_{\beta} G$.

3. CHAMBER IS A DOUBLING METRIC MEASURE SPACE

Theorem 3.1. *Let Γ be a chamber. For all $x \in \Gamma$ and for all $r > 0$, we have*

$$w(B(x, r) \cap \Gamma) \geq c_{\kappa} w(B(x, r)).$$

In particular, $(\Gamma, |\cdot|, w(x)dx)$ is a doubling metric measure space.

Notation 3.2. *Let $\theta : 2^{N_1} \rightarrow \mathbb{R}_+$. Set*

$$w_{\theta}(x) = \prod_{A \subset \{1, \dots, N_1\}} \left(\sum_{k \in A} x_k \right)^{\theta_A}, \quad x \in \mathbb{R}_+^{N_1}.$$

Lemma 3.3. *For every $\theta : 2^{N_1} \rightarrow \mathbb{R}_+$ and for every $x \in \mathbb{R}_+^{N_1} \times \mathbb{R}^{N_2}$, we have*

$$\int_{B(x, r)} w_{\theta}(y_1, \dots, y_{N_1}) dy \leq c_{N, \theta}^{(1)} r^N w_{\theta}((x_1, \dots, x_{N_1}) + r \mathbf{1}_{N_1}).$$

Proof. Obviously,

$$B(x, r) \subset \left(\times_{k=1}^{N_1} [(x_k - r)_+, x_k + r] \right) \times B((x_{N_1+1}, \dots, x_N), r).$$

Thus,

$$\begin{aligned} & \int_{B(x, r)} w_{\theta}(y_1, \dots, y_{N_1}) dy \leq \\ & \leq \int_{\left(\times_{k=1}^{N_1} [(x_k - r)_+, x_k + r] \right) \times B((x_{N_1+1}, \dots, x_N), r)} w_{\theta}(y_1, \dots, y_{N_1}) dy = \\ & = \int_{\times_{k=1}^{N_1} [(x_k - r)_+, x_k + r]} w_{\theta}(u) du \times \text{Vol}(\mathbb{B}_{N_2}) r^{N_2} \leq \\ & \leq \int_{\times_{k=1}^{N_1} [0, x_k + r]} w_{\theta}((x_1, \dots, x_{N_1}) + r \mathbf{1}_{N_1}) du \times \text{Vol}(\mathbb{B}_{N_2}) r^{N_2} = \\ & = w_{\theta}((x_1, \dots, x_{N_1}) + r \mathbf{1}_{N_1}) \times \prod_{k=1}^{N_1} (x_k + r - (x_k - r)_+) \times \text{Vol}(\mathbb{B}_{N_2}) r^{N_2} \leq \\ & \leq w_{\theta}((x_1, \dots, x_{N_1}) + r \mathbf{1}_{N_1}) \times (2r)^{N_1} \times \text{Vol}(\mathbb{B}_{N_2}) r^{N_2}. \end{aligned}$$

□

Lemma 3.4. *For every $\theta : 2^{N_1} \rightarrow \mathbb{R}_+$ and for every $x \in \mathbb{R}_+^{N_1} \times \mathbb{R}^{N_2}$, we have*

$$\int_{B(x, r) \cap (\mathbb{R}_+^{N_1} \times \mathbb{R}^{N_2})} w_{\theta}(y_1, \dots, y_{N_1}) dy \geq c_{N, \theta}^{(2)} r^N w_{\theta}((x_1, \dots, x_{N_1}) + r \mathbf{1}_{N_1}).$$

Proof. Obviously,

$$x + [\frac{r}{2N}, \frac{r}{N}]^N \subset B(x, r) \cap (\mathbb{R}_+^{N_1} \times \mathbb{R}^{N_2}).$$

It follows that

$$\begin{aligned} & \int_{B(x, r) \cap (\mathbb{R}_+^{N_1} \times \mathbb{R}^{N_2})} w_\theta(y_1, \dots, y_{N_1}) dy \geq \\ & \geq \int_{x + [\frac{r}{2N}, \frac{r}{N}]^N} w_\theta(y_1, \dots, y_{N_1}) dy = \\ & = \int_{(x_1, \dots, x_{N_1}) + [\frac{r}{2N}, \frac{r}{N}]^{N_1}} w_\theta(u) du \times (\frac{r}{2N})^{N_2} \geq \\ & \geq \int_{(x_1, \dots, x_{N_1}) + [\frac{r}{2N}, \frac{r}{N}]^{N_1}} w_\theta((x_1, \dots, x_{N_1}) + \frac{r}{2N} \mathbf{1}_{N_1}) du \times (\frac{r}{2N})^{N_2} = \\ & = w_\theta((x_1, \dots, x_{N_1}) + \frac{r}{2N} \mathbf{1}_{N_1}) \times (\frac{r}{2N})^N \geq \\ & \geq w_\theta((x_1, \dots, x_{N_1}) + r \mathbf{1}_{N_1}) \times (\frac{r}{2N})^N \times (2N)^{-\sum_{A \subset \{1, \dots, N_1\}} \theta_A}. \end{aligned}$$

□

Lemma 3.5. *There exists $\theta : 2^{N_1} \rightarrow \mathbb{R}_+$ such that*

$$c_\kappa^{-1} w_\theta(x_1, \dots, x_{N_1}) \leq w(\Phi^{-1}(x)) \leq c_\kappa w_\theta(x_1, \dots, x_{N_1}), \quad x \in \mathbb{R}_+^{N_1} \times \mathbb{R}^{N_2}.$$

Proof. For every $x \in \mathbb{R}_+^{N_1} \times \mathbb{R}^{N_2}$ and for every $v \in R$, we have

$$\langle \Phi^{-1}(x), v \rangle = \langle x, (\Phi^{-1})^*(v) \rangle.$$

The latter expression has the same sign (positive or negative) on the whole cone $\mathbb{R}_+^{N_1} \times \mathbb{R}^{N_2}$. Hence,

$$((\Phi^{-1})^*(v))_k = \begin{cases} \text{something positive,} & 1 \leq k \leq N_1 \\ 0, & k > N_1 \end{cases}$$

Let A_v be the set of those $1 \leq k \leq N_1$ with $((\Phi^{-1})^*(v))_l$ being strictly positive. We, obviously, have

$$c_{\Phi, v}^{-1} \sum_{k \in A_v} x_k \leq |\langle \Phi^{-1}(x), v \rangle| \leq c_{\Phi, v} \sum_{k \in A_v} x_k, \quad x \in \mathbb{R}_+^{N_1} \times \mathbb{R}^{N_2}.$$

Thus,

$$\prod_{v \in R} c_{\Phi, v}^{-\kappa(v)} \left(\sum_{k \in A_v} x_k \right)^{\kappa(v)} \leq w(x) \leq \prod_{v \in R} c_{\Phi, v}^{\kappa(v)} \left(\sum_{k \in A_v} x_k \right)^{\kappa(v)}, \quad x \in \mathbb{R}_+^{N_1} \times \mathbb{R}^{N_2}.$$

This completes the proof. □

Proof of Theorem 3.1. The first assertion follows by combining Lemmas 3.3, 3.4 and 3.5. The second assertion is an obvious consequence of the first one. □

4. ROCHBERG-SEMMES BASED ESTIMATE

Theorem 4.1. *Let $N \geq 2$, $1 \leq N_1 \leq N$ and $N_2 = N - N_1$. Let $\Omega = \mathbb{R}_+^{N_1} \times \mathbb{R}^{N_2}$. Let $\mathbf{w} \in A_2(\Omega)$ be such that*

$$\sup_{x \in Q} \mathbf{w}(x) \leq c_w \frac{\mathbf{w}(Q)}{m(Q)}, \quad Q \in \mathbb{D}, \quad Q \subset \Omega.$$

Suppose that $K : \Omega \times \Omega \rightarrow \mathbb{C}$ satisfies

$$|\partial_x^\alpha \partial_y^\beta K|(x, y) \leq c_{K, \alpha, \beta} \frac{|x - y|^{1 - |\alpha|_1 - |\beta|_1}}{\mathbf{w}(B(x, |x - y|))}, \quad x, y \in \Omega, \quad \alpha, \beta \in \mathbb{Z}_+^N.$$

Let V be an integral operator on $L_2(\Omega, \mathbf{w}(x)dx)$ with integral kernel K . Suppose $N > 2s$. For every $f \in C_c^\infty(\mathbb{R}^N)$, we have

$$\|M_f V\|_{\mathcal{L}_{N, \infty}(L_2(\Omega, \mathbf{w}(x)dx))} \leq c_{K, \mathbf{w}} \|f\|_{L_N(\Omega)}.$$

Lemma 4.2. *There exists a smooth function ψ supported on $[-1, 1]^N \setminus (-\frac{1}{4}, \frac{1}{4})^N$ such that*

$$1 = \sum_{Q \in \mathbb{D}} \chi_Q(x) \chi_{3Q}(y) \psi\left(\frac{x - y}{\ell(Q)}\right).$$

Proof. Choose $\rho \in C_c^\infty(\mathbb{R}^N)$ such that $\rho = 1$ on $(-\frac{1}{2}, \frac{1}{2})^N$ and $\rho = 0$ outside $(-1, 1)^N$. Set $\psi(z) = \rho(z) - \rho(2z)$. Clearly,

$$\text{supp}(\psi) \subset \left\{ z \in \mathbb{R}^N : \frac{1}{4} \leq |z|_\infty \leq 1 \right\}.$$

For every $z \neq 0$, we have

$$(3) \quad \sum_{j \in \mathbb{Z}} \psi(2^j z) = 1.$$

For every x , there exists a unique $Q \in \mathbb{D}_j$ containing x . For this Q , we have $\ell(Q) = 2^{-j}$. We have

$$\begin{aligned} 1 &= \sum_{j \in \mathbb{Z}} \psi(2^j(x - y)) = \sum_{j \in \mathbb{Z}} \left(\sum_{Q \in \mathbb{D}_j} \chi_Q(x) \right) \psi(2^j(x - y)) = \\ &= \sum_{j \in \mathbb{Z}} \sum_{Q \in \mathbb{D}_j} \chi_Q(x) \psi\left(\frac{x - y}{\ell(Q)}\right) = \sum_{Q \in \mathbb{D}} \chi_Q(x) \psi\left(\frac{x - y}{\ell(Q)}\right). \end{aligned}$$

If $x \in Q$ and $\psi(\frac{x-y}{\ell(Q)}) \neq 0$, then

$$x \in Q, \quad |x - y|_\infty \leq \ell(Q).$$

We obviously have $|x - c_Q|_\infty \leq \frac{1}{2}\ell(Q)$ and $|x - y|_\infty \leq \ell(Q)$. Thus, $|y - c_Q|_\infty \leq \frac{3}{2}\ell(Q)$. In other words, $y \in 3Q$. Thus,

$$\chi_Q(x) \psi\left(\frac{x - y}{\ell(Q)}\right) = \chi_Q(x) \chi_{3Q}(y) \psi\left(\frac{x - y}{\ell(Q)}\right).$$

This completes the proof. $\square$

Lemma 4.3. *Let $r > 2$. In the setting of Theorem 4.1, we have*

$$f(x)K(x, y) = \sum_{k, l \in \mathbb{Z}^N} (1 + |k|^2 + |l|^2)^{-N-1} \sum_{\substack{Q \in \mathbb{D} \\ Q \subset \Omega}} e_{k, l, Q}(x) f_{k, l, Q}(y),$$

where

(i) $e_{k,l,Q}$ is supported in Q and $f_{k,l,Q}$ is supported in $3Q$ for every $Q \in \mathbb{D}$ with $Q \subset \Omega$;

(ii) for every $k, l \in \mathbb{Z}^N$ and for every $Q \in \mathbb{D}$ with $Q \subset \Omega$, we have

$$\|e_{k,l,Q}\|_{L_r(\mathbb{R}^N, \mathbf{w}(x)dx)} = \|f\|_{L_r(Q, \mathbf{w}(x)dx)};$$

(iii) for every $k, l \in \mathbb{Z}^N$ and for every $Q \in \mathbb{D}$ with $Q \subset \Omega$, we have

$$\|f_{k,l,Q}\|_{L_\infty(\mathbb{R}^N)} \leq \frac{c_{K,\theta} \cdot \ell(Q)}{\mathbf{w}(Q)};$$

Proof. For every $x, y \in \Omega$, we have

$$\begin{aligned} K(x, y) &= \sum_{Q \in \mathbb{D}} \chi_Q(x) \chi_{3Q}(y) K(x, y) \psi\left(\frac{x-y}{\ell(Q)}\right) = \\ &= \sum_{\substack{Q \in \mathbb{D} \\ Q \subset \Omega}} \chi_Q(x) \chi_{3Q}(y) K(x, y) \psi\left(\frac{x-y}{\ell(Q)}\right) = \end{aligned}$$

For every $Q \in \mathbb{D}$ with $3Q \subset \Omega$, define the map

$$K_Q : \left[-\frac{1}{2}, \frac{1}{2}\right]^N \times \left[-\frac{3}{2}, \frac{3}{2}\right]^N \rightarrow \mathbb{C}$$

by setting

$$K_Q(a, b) = K(c_Q + a\ell(Q), c_Q + b\ell(Q))\psi(a-b), \quad a \in \left[-\frac{1}{2}, \frac{1}{2}\right]^N, \quad b \in \left[-\frac{3}{2}, \frac{3}{2}\right]^N.$$

If $Q \in \mathbb{D}$ is such that $3Q \not\subset \Omega$, then the definition of K_Q is the same, a also runs through $[-\frac{1}{2}, \frac{1}{2}]^N$, but the parallelepiped b runs through is slightly smaller. **to co-authors: if you wish, specify the domain of b also.** In what follows, we lighten the notations by pretending that b is always defined on $[-\frac{3}{2}, \frac{3}{2}]^N$.

It follows from the assumption on K that

$$\|K_Q\|_{C^{2N+2}([- \frac{3}{2}, \frac{3}{2}]^N \times [- \frac{3}{2}, \frac{3}{2}]^N)} \leq \frac{c'_{K,\theta} \cdot \ell(Q)}{\mathbf{w}(Q)}.$$

Hence, there exists an extension $K_Q : \mathbb{T}^N \times \mathbb{T}^N \rightarrow \mathbb{C}$ such that

$$\|K_Q\|_{C^{2N+2}(\mathbb{T}^N \times \mathbb{T}^N)} \leq \frac{c''_{K,\theta} \cdot \ell(Q)}{\mathbf{w}(Q)}.$$

In particular,

$$\|K_Q\|_{W^{2N+2,2}(\mathbb{T}^N \times \mathbb{T}^N)} \leq \frac{c_{K,\theta} \cdot \ell(Q)}{\mathbf{w}(Q)}.$$

We may, therefore, write

$$K_Q(a, b) = \sum_{k,l \in \mathbb{Z}^N} \text{coef}_{k,l,Q} e^{i\langle k,a \rangle} e^{i\langle l,b \rangle}, \quad a, b \in \mathbb{T}^N,$$

where

$$\sum_{k,l \in \mathbb{Z}^N} (1 + |k|^2 + |l|^2)^{2N+2} |\text{coef}_{k,l,Q}|^2 \leq \left(\frac{c_{K,\theta} \cdot \ell(Q)}{\mathbf{w}(Q)} \right)^2.$$

In particular,

$$(1 + |k|^2 + |l|^2)^{N+1} \cdot |\text{coef}_{k,l,Q}| \leq \frac{c_{K,\theta} \cdot \ell(Q)}{\mathbf{w}(Q)}, \quad k, l \in \mathbb{Z}^N.$$

We now set

$$e_{k,l,Q}(x) = f(x)\chi_Q(x)e^{i\langle k, \frac{x-c_Q}{\ell(Q)} \rangle},$$

$$f_{k,l,Q}(y) = (1 + |k|^2 + |l|^2)^{N+1} \text{coef}_{k,l,Q} \cdot \chi_{3Q}(y)e^{i\langle l, \frac{y-c_Q}{\ell(Q)} \rangle}.$$

This allows us to write

$$f(x)K(x, y) = \sum_{k,l \in \mathbb{Z}^N} (1 + |k|^2 + |l|^2)^{-N-1} \sum_{\substack{Q \in \mathbb{D} \\ Q \subset \Omega}} e_{k,l,Q}(x) f_{k,l,Q}(y).$$

Obviously,

$$\begin{aligned} \text{supp}(e_{k,l,Q}) &\subset Q, \quad \text{supp}(f_{k,l,Q}) \subset 3Q, \\ \|e_{k,l,Q}\|_{L_r(\mathbb{R}^N, \mathbf{w}(x)dx)} &= \|f\|_{L_r(Q, \mathbf{w}(x)dx)}, \\ \|f_{k,l,Q}\|_{L_\infty(\mathbb{R}^N)} &= (1 + |k|^2 + |l|^2)^{N+1} |\text{coef}_{k,l,Q}| \leq \frac{c_{K,\theta} \cdot \ell(Q)}{\mathbf{w}(Q)}. \end{aligned}$$

□

Lemma 4.4. *Let $r > 2$. In the setting of Theorem 4.1, we have*

$$\|M_f V\|_{\mathcal{L}_{N,\infty}(L_2(\Omega, \mathbf{w}(x)dx))} \leq c_{K,\mathbf{w},r} \left\| \left\{ m(Q)^{\frac{1}{N}-\frac{1}{r}} \|f\|_{L_r(Q)} \right\}_{\substack{Q \in \mathbb{D} \\ Q \subset \Omega}} \right\|_{N,\infty}.$$

Proof. Let $A_{k,l}$ be the integral operator with integral kernel given by the formula

$$(x, y) \rightarrow \sum_{\substack{Q \in \mathbb{D} \\ Q \subset \Omega}} e_{k,l,Q}(x) f_{k,l,Q}(y), \quad x, y \in \Omega, \quad k, l \in \mathbb{Z}^N.$$

By Lemma 4.3, we have

$$M_f V = \sum_{k,l \in \mathbb{Z}^N} (1 + |k|^2 + |l|^2)^{-N-1} A_{k,l}.$$

By the triangle inequality,

$$\begin{aligned} &\|M_f V\|_{\mathcal{L}_{N,\infty}(L_2(\Omega, w_\theta(x)dx))} \leq \\ &\leq \frac{N}{N-1} \sum_{k,l \in \mathbb{Z}^N} (1 + |k|^2 + |l|^2)^{-N-1} \|A_{k,l}\|_{\mathcal{L}_{N,\infty}(L_2(\Omega, w_\theta(x)dx))}. \end{aligned}$$

By Lemma 4.3, the sequences

$$\left\{ \mathbf{w}(Q)^{\frac{1}{r}-\frac{1}{2}} \frac{e_{k,l,Q}}{\|f\|_{L_r(Q, \mathbf{w}(x)dx)}} \right\}_{\substack{Q \in \mathbb{D} \\ Q \subset \Omega}}, \quad \left\{ \frac{c_{K,\theta}^{-1} \mathbf{w}(Q)^{\frac{1}{2}}}{\ell(Q)} f_{k,l,Q} \right\}_{\substack{Q \in \mathbb{D} \\ Q \subset \Omega}}$$

are $(r, 1)$ -NWO and, respectively, $(\infty, 3)$ -NWO in the sense of Definition 2.4. Using Lemma 2.5, we conclude that

$$\begin{aligned} &\|A_{k,l}\|_{\mathcal{L}_{N,\infty}(L_2(\Omega, \mathbf{w}(x)dx))} \leq \\ &\leq c_{\theta,r,1,\infty,3,N} c_{K,\theta} \left\| \left\{ \ell(Q) \mathbf{w}(Q)^{-\frac{1}{r}} \|f\|_{L_r(Q, \mathbf{w}(x)dx)} \right\}_{\substack{Q \in \mathbb{D} \\ Q \subset \Omega}} \right\|_{N,\infty}. \end{aligned}$$

Therefore,

$$\begin{aligned} &\|M_f V\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}_+^N, \mathbf{w}(x)dx))} \leq \\ &\leq \frac{N}{N-1} c_{\theta,r,1,\infty,3,N} c_{K,\theta} \sum_{k,l \in \mathbb{Z}^N} (1 + |k|^2 + |l|^2)^{-N-1} \times \end{aligned}$$

$$\times \left\| \left\{ \ell(Q) \mathbf{w}(Q)^{-\frac{1}{r}} \|f\|_{L_r(Q, \mathbf{w}(x)dx)} \right\}_{Q \in \mathbb{D}} \right\|_{N, \infty}.$$

Obviously,

$$\mathbf{w}(Q)^{-\frac{1}{r}} \|f\|_{L_r(Q, \mathbf{w}(x)dx)} \leq \left(\frac{1}{\mathbf{w}(Q)} \cdot \sup_{x \in Q} \mathbf{w}(x) \right)^{\frac{1}{r}} \times \|f\|_{L_r(Q)}, \quad Q \in \mathbb{D}, \quad Q \subset \Omega.$$

Combining the last two estimates, we complete the proof. $\square$

Proof of Theorem 4.1. Take $r \in (2, N)$. The assertion follows by combining Lemmas 4.4 and A.1. $\square$

5. PROOF OF SPECIAL CBIKEL ESTIMATE IN DUNKL SETTING

Lemma 5.1. *For every pair of multi-indices $\alpha, \beta \in \mathbb{Z}_+^N$, for every $t > 0$, we have*

$$|(\partial_x^\alpha \partial_y^\beta h_t)(x, y)| \leq c_{\kappa, \alpha, \beta} t^{-\frac{|\alpha|_1 + |\beta|_1}{2}} w(B(x, t^{\frac{1}{2}}))^{-1} \exp(-\frac{c_\kappa}{t} d(x, y)^2).$$

Proof. This is (slightly weakened form of) Theorem 3.1 in [2]. $\square$

Lemma 5.2. *Suppose $s + N > 2$. We have*

$$\int_0^\infty t^{-\frac{s}{2}} w(B(x, t^{\frac{1}{2}}))^{-1} \exp(-\frac{c_\kappa}{t} d(x, y)^2) dt \leq c_{\kappa, s} w(B(x, d(x, y)))^{-1} \times d(x, y)^{2-s}.$$

Proof. By (2),

$$w(B(x, t^{\frac{1}{2}}))^{-1} \leq c_\kappa^{(1)} w(B(x, d(x, y)))^{-1} \times \begin{cases} \left(\frac{d(x, y)}{t^{\frac{1}{2}}} \right)^N, & 0 < t \leq d(x, y)^2, \\ \left(\frac{d(x, y)}{t^{\frac{1}{2}}} \right)^N, & t \geq d(x, y)^2. \end{cases}$$

Therefore,

$$\begin{aligned} & \int_0^\infty t^{-\frac{s}{2}} w(B(x, t^{\frac{1}{2}}))^{-1} \exp(-\frac{c_\kappa}{t} d(x, y)^2) dt \leq \\ & \leq c_\kappa^{(1)} w(B(x, d(x, y)))^{-1} \times \int_0^{d(x, y)^2} t^{-\frac{s}{2}} \left(\frac{d(x, y)}{t^{\frac{1}{2}}} \right)^N \exp(-\frac{c_\kappa}{t} d(x, y)^2) dt + \\ & \quad + c_\kappa^{(1)} w(B(x, d(x, y)))^{-1} \times \int_{d(x, y)^2}^\infty t^{-\frac{s}{2}} \left(\frac{d(x, y)}{t^{\frac{1}{2}}} \right)^N dt = \\ & = c_\kappa^{(1)} w(B(x, d(x, y)))^{-1} \times d(x, y)^{2-s} \times \int_0^1 t^{-\frac{s+N}{2}} \exp(-\frac{c_\kappa}{t}) dt + \\ & \quad + c_\kappa^{(1)} w(B(x, d(x, y)))^{-1} \times d(x, y)^{2-s} \times \int_1^\infty t^{-\frac{s+N}{2}} dt. \end{aligned}$$

$\square$

The following lemma establishes derivative estimates for the integral kernels of Dunkl-Riesz transforms.

Lemma 5.3. *Let $N \geq 2$. For every pair of multi-indices $\alpha, \beta \in \mathbb{Z}_+^n$, we have*

$$|(\partial_x^\alpha \partial_y^\beta K_{\Delta_{\text{Dunkl}}^{-\frac{1}{2}}})(x, y)| \leq c_{\kappa, \alpha, \beta} \frac{d(x, y)^{1-|\alpha|_1-|\beta|_1}}{w(B(x, d(x, y)))}, \quad x, y \in \mathbb{R}^N.$$

Proof. We have

$$\Delta_{\text{Dunkl}}^{-\frac{1}{2}} = \pi^{-\frac{1}{2}} \int_0^\infty e^{-t\Delta_{\text{Dunkl}}} \frac{dt}{t^{\frac{1}{2}}}.$$

Hence,

$$K_{\Delta_{\text{Dunkl}}^{-\frac{1}{2}}}(x, y) = \pi^{-\frac{1}{2}} \int_0^\infty h_t(x, y) \frac{dt}{t^{\frac{1}{2}}}.$$

Using Lemma 5.1, we estimate

$$\begin{aligned} & |(\partial_x^\alpha \partial_y^\beta K_{R_{\text{Dunkl},j}})(x, y)| \leq \\ & \leq \frac{c_{\kappa, \alpha, \beta}}{\pi^{\frac{1}{2}}} \int_0^\infty t^{-\frac{|\alpha|_1 + |\beta|_1 + 1}{2}} w(B(x, t^{\frac{1}{2}}))^{-1} \exp(-\frac{c_\kappa}{t} d(x, y)^2) dt. \end{aligned}$$

Applying Lemma 5.2 with $s = 1 + |\alpha|_1 + |\beta|_1$, we conclude

$$\begin{aligned} & |(\partial_x^\alpha \partial_y^\beta K_{\Delta_{\text{Dunkl}}^{-\frac{1}{2}}})(x, y)| \leq \\ & \leq \pi^{-\frac{1}{2}} c_{\kappa, \alpha, \beta} \times c_{\kappa, |\alpha|_1 + |\beta|_1 + 1} w(B(x, d(x, y)))^{-1} \times d(x, y)^{1 - |\alpha|_1 - |\beta|_1}. \end{aligned}$$

The assertion follows immediately. $\square$

Lemma 5.4. *For every $Q \in \mathbb{D}$ with $Q \subset \Omega$, we have*

$$\sup_{x \in Q} w_\theta(x) \leq c_\theta \frac{w_\theta(Q)}{m(Q)}.$$

Proof. We have

$$w(Q) \geq c_\theta \prod_{A \subset \{1, \dots, N\}} (\ell_Q + \sum_{k \in A} (c_Q)_k)^{\theta_A}.$$

On the other hand, if $x \in Q$, then $|x - c_Q|_\infty \leq \frac{1}{2} \ell(Q)$ and, therefore,

$$\sum_{k \in A} x_k \leq \frac{|A|}{2} \ell_Q + \sum_{k \in A} (c_Q)_k \leq N(\ell_Q + \sum_{k \in A} (c_Q)_k).$$

Thus,

$$w(x) = \prod_{A \subset \{1, \dots, N\}} (\sum_{k \in A} x_k)^{\theta_A} \leq N^{\sum_{A \subset \{1, \dots, N\}} \theta_A} \prod_{A \subset \{1, \dots, N\}} (\ell_Q + \sum_{k \in A} (c_Q)_k)^{\theta_A}.$$

This completes the proof. $\square$

Theorem 5.5. *Let $N \geq 3$. We have*

$$\|M_f \Delta_{\text{Dunkl}}^{-\frac{1}{2}}\|_{\mathcal{L}_{\frac{N}{s}, \infty}(L_2(\mathbb{R}^N, w(x)dx))} \leq c_\kappa \|f\|_{L_N(\mathbb{R}^N)}.$$

Proof. It suffices to estimate, for every chamber Γ and for every $g \in G$, the operator $M_{\chi_\Gamma \cdot f} \Delta_{\text{Dunkl}}^{-\frac{1}{2}} M_{\chi_{g\Gamma}}$. In other words, it suffices to estimate an operator with integral kernel

$$f(x) K_{\text{Dunkl}^{-\frac{1}{2}}}(x, y), \quad x, y \in \Gamma, \quad y \in g\Gamma,$$

on the Hilbert space $L_2(\mathbb{R}^N, w(x)dx)$. In other words, it suffices to estimate an operator with integral kernel

$$f(x) K_{\text{Dunkl}^{-\frac{1}{2}}}(x, gy), \quad x, y \in \Gamma,$$

on the Hilbert space $L_2(\Gamma, w(x)dx)$.

Let $N_1 = \dim(\text{span}(R))$, $N_2 = N - N_1$ and let $\Omega = \mathbb{R}_+^{N_1} \times \mathbb{R}^{N_2}$. For a concrete chamber Γ , let $\Phi : \mathbb{R}^N \rightarrow \mathbb{R}^N$ be a linear isomorphism given in Theorem 2.2 so that $\Phi(\Gamma) = \Omega$.

It suffices to estimate an operator with integral kernel

$$f(\Phi^{-1}x)K_{\text{Dunkl}^{-\frac{1}{2}}}(\Phi^{-1}x, g\Phi^{-1}y), \quad x, y \in \Omega,$$

on the Hilbert space $L_2(\Omega, w(\Phi^{-1}x)dx)$. Using Lemma 3.5, it suffices to estimate an operator with integral kernel

$$f(\Phi^{-1}x)K_{\text{Dunkl}^{-\frac{1}{2}}}(\Phi^{-1}x, g\Phi^{-1}y), \quad x, y \in \Omega,$$

on the Hilbert space $L_2(\Omega, w_\theta(x)dx)$.

By Lemma 5.3,

$$\left| \partial_x^\alpha \partial_y^\beta K_{\text{Dunkl}^{-\frac{1}{2}}}(\Phi^{-1}x, g\Phi^{-1}y) \right| \leq c_{\alpha, \beta, \kappa} \frac{|x - y|^{1 - |\alpha|_1 - |\beta|_1}}{w_\theta(B(x, |x - y|))}, \quad x, y \in \Omega.$$

The other assumptions in Theorem 4.1 follow from Lemma 5.4, Lemma 3.3 and Lemma 3.4. The assertion follows now from Theorem 4.1. $\square$

6. PROOF OF THEOREM 1.1 (i)

Lemma 6.1. *We have*

$$[T_{e_j}, M_f] = M_{\partial_j f} + \frac{1}{2} \sum_{v \in R} v_j \kappa(v) M_{\delta_v f} U_v,$$

where

$$(\delta_v f)(x) = \frac{f(x) - (U_v f)(x)}{\langle v, x \rangle}, \quad v \in R, \quad x \in \mathbb{R}^N.$$

Proof. Obvious. $\square$

Lemma 6.2. *We have*

$$\|\delta_v f\|_{L_N(\mathbb{R}^N)} \leq c_N \|f\|_{\dot{W}^{1, N}(\mathbb{R}^N)}.$$

Proof. Without loss of generality, $v = e_N$. Set

$$(D_t h)(x) = h(x_1, \dots, x_{N-1}, tx_N).$$

We have

$$\delta_v f = \int_{-1}^1 D_t(\partial_N f) dt.$$

Thus,

$$\begin{aligned} \|\delta_v f\|_{L_N(\mathbb{R}^N)} &\leq \int_{-1}^1 \|D_t(\partial_N f)\|_{L_N(\mathbb{R}^N)} dt = \\ &= \int_{-1}^1 |t|^{-\frac{1}{N}} \|\partial_N f\|_{L_N(\mathbb{R}^N)} dt = \frac{2N}{N-1} \|\partial_N f\|_{L_N(\mathbb{R}^N)}. \end{aligned}$$

$\square$

Lemma 6.3. *For every $N > 1$, we have*

$$\|[B, A]B^{-1}\|_{N, \infty} \leq c_N \|B^{-1}[B^2, A]B^{-1}\|_{N, \infty}.$$

Proof of Theorem 1.1 (i). We prove the assertion under *a priori* restriction $f \in C_c^\infty(\mathbb{R}^N)$. Passing to the case of $f \in \dot{W}^{1,N}(\mathbb{R}^N)$ is standard (see e.g. [refer to our papers](#)).

We have

$$\begin{aligned} [R_{\text{Dunkl},j}, M_f] &= [T_{e_j} \Delta_{\text{Dunkl}}^{-\frac{1}{2}}, M_f] = [T_{e_j}, M_f] \Delta_{\text{Dunkl}}^{-\frac{1}{2}} + T_{e_j} [\Delta_{\text{Dunkl}}^{-\frac{1}{2}}, M_f] = \\ &= [T_{e_j}, M_f] \Delta_{\text{Dunkl}}^{-\frac{1}{2}} - R_{\text{Dunkl},j} [\Delta_{\text{Dunkl}}^{\frac{1}{2}}, M_f] \Delta_{\text{Dunkl}}^{-\frac{1}{2}}. \end{aligned}$$

By triangle inequality,

$$\begin{aligned} \|[R_{\text{Dunkl},j}, M_f]\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))} &\leq \\ &\leq \frac{N}{N-1} \|[T_{e_j}, M_f] \Delta_{\text{Dunkl}}^{-\frac{1}{2}}\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))} + \\ &+ \frac{N}{N-1} \|[\Delta_{\text{Dunkl}}^{\frac{1}{2}}, M_f] \Delta_{\text{Dunkl}}^{-\frac{1}{2}}\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))}. \end{aligned}$$

Using Lemma 6.3, we write

$$\begin{aligned} \|[R_{\text{Dunkl},j}, M_f]\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))} &\leq \\ &\leq \frac{N}{N-1} \|[T_{e_j}, M_f] \Delta_{\text{Dunkl}}^{-\frac{1}{2}}\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))} + \\ &+ \frac{N}{N-1} c_N \|\Delta_{\text{Dunkl}}^{-\frac{1}{2}} [\Delta_{\text{Dunkl}}, M_f] \Delta_{\text{Dunkl}}^{-\frac{1}{2}}\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))}. \end{aligned}$$

Note that

$$\begin{aligned} \Delta_{\text{Dunkl}}^{-\frac{1}{2}} [\Delta_{\text{Dunkl}}, M_f] \Delta_{\text{Dunkl}}^{-\frac{1}{2}} &= \\ &= \sum_{k=1}^N \Delta_{\text{Dunkl}}^{-\frac{1}{2}} [T_{e_k}, M_f] R_{\text{Dunkl},k} + R_{\text{Dunkl},k} [T_{e_k}, M_f] \Delta_{\text{Dunkl}}^{-\frac{1}{2}}. \end{aligned}$$

By triangle inequality,

$$\begin{aligned} \|\Delta_{\text{Dunkl}}^{-\frac{1}{2}} [\Delta_{\text{Dunkl}}, M_f] \Delta_{\text{Dunkl}}^{-\frac{1}{2}}\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))} &\leq \\ &\leq \frac{N}{N-1} \sum_{k=1}^N \|\Delta_{\text{Dunkl}}^{-\frac{1}{2}} [T_{e_k}, M_f]\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))} + \\ &+ \frac{N}{N-1} \sum_{k=1}^N \|[T_{e_k}, M_f] \Delta_{\text{Dunkl}}^{-\frac{1}{2}}\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))}. \end{aligned}$$

Combining the estimates in the last two paragraphs, we obtain

$$\begin{aligned} \|[R_{\text{Dunkl},j}, M_f]\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))} &\leq \\ &\leq c'_N \sum_{k=1}^N \|\Delta_{\text{Dunkl}}^{-\frac{1}{2}} [T_{e_k}, M_f]\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))} + \\ &+ c'_N \sum_{k=1}^N \|[T_{e_k}, M_f] \Delta_{\text{Dunkl}}^{-\frac{1}{2}}\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))}. \end{aligned}$$

Using Lemma 6.1, the fact that Δ_{Dunkl} commutes with O_v and the triangle inequality, we obtain

$$\begin{aligned} \|[T_{e_k}, M_f] \Delta_{\text{Dunkl}}^{-\frac{1}{2}}\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))} &\leq \\ &\leq \frac{N}{N-1} \|M_{\partial_k} f \Delta_{\text{Dunkl}}^{-\frac{1}{2}}\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))} + \end{aligned}$$

$$+ \frac{N}{2(N-1)} \sum_{v \in R} |v| \kappa(v) \|M_{\delta_v f} \Delta_{\text{Dunkl}}^{-\frac{1}{2}}\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))}.$$

Thus,

$$\begin{aligned} & \| [R_{\text{Dunkl},j}, M_f] \|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))} \leq \\ & \leq c_N'' \sum_{k=1}^N \|M_{\partial_k f} \Delta_{\text{Dunkl}}^{-\frac{1}{2}}\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))} + \\ & + c_\kappa \sum_{v \in R} \|M_{\delta_v f} \Delta_{\text{Dunkl}}^{-\frac{1}{2}}\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))}. \end{aligned}$$

The assertion follows now from Theorem 5.5 and Lemma 6.2. $\square$

7. PROOF OF THEOREM 1.1 (ii)

Lemma 7.1. *Let $1 \leq j \leq N$. If $x_j \geq y_j$, then*

$$K_{\text{Dunkl},j}(x, y) \geq c_\kappa \frac{x_j - y_j}{|x - y|} \cdot \frac{1}{w(B(x, |x - y|))}.$$

If $x_j \leq y_j$, then

$$K_{\text{Dunkl},j}(x, y) \leq c_\kappa \frac{x_j - y_j}{|x - y|} \cdot \frac{1}{w(B(x, |x - y|))}.$$

Proof. The proof is identical to that in [6, Theorem 1.2]. $\square$

Lemma 7.2. *Let $Q \in \mathbb{D}$. We have*

$$K_j(x, y) \geq \frac{c_\kappa}{w(Q)}, \quad x \in 3Q, \quad y \in Q + 3s(Q)e_j.$$

Proof. The assertion follows immediately from Lemma 7.1. $\square$

Lemma 7.3. *Let real-valued $f \in L_1^{\text{loc}}(\mathbb{R}^N)$ be such that the commutator $[R_{\text{Dunkl},j}, M_f]$ is bounded on $L_2(\mathbb{R}^N, w(x)dx)$. For every $Q \in \mathbb{D}$, there exist ξ_Q and η_Q supported in $7Q$ such that $\|\xi_Q\|_\infty, \|\eta_Q\|_\infty \leq m(Q)^{-\frac{1}{2}}$ and such that*

$$\frac{1}{w(3Q)} \int_{3Q} |f(y) - f_{3Q}| w(y) dy \leq c_\kappa |\langle [R_{\text{Dunkl},j}, M_f] \xi_Q, \eta_Q \rangle_{L_2(\mathbb{R}^N, w(x)dx)}|.$$

Proof. Let α be the median of f on $3Q$ and β be the median of f on $Q + 3s(Q)e_j$. Assume for definiteness $\alpha \geq \beta$. Set

$$\begin{aligned} A_1 &= \{x \in 3Q : f(x) \geq \alpha\}, \quad B_1 = \{x \in Q + 3s(Q)e_j : f(x) \leq \beta\}, \\ A_2 &= \{x \in 3Q : f(x) \leq \beta\}, \quad B_2 = \{x \in Q + 3s(Q)e_j : f(x) \geq \alpha\}, \\ \xi_{Q,s} &= w(Q)^{-\frac{1}{2}} \chi_{A_s}, \quad \eta_{Q,s} = w(Q)^{-\frac{1}{2}} \chi_{B_s}, \quad s = 1, 2. \end{aligned}$$

We have

$$\begin{aligned} & \langle [R_{\text{Dunkl},j}, M_f] \xi_{Q,s}, \eta_{Q,s} \rangle_{L_2(\mathbb{R}^N, w(x)dx)} = \\ & = \frac{1}{w(Q)} \int_{B_s \times A_s} K_j(x, y) (f(y) - f(x)) w(x) w(y) dx dy. \end{aligned}$$

If $x \in B_1$ and $y \in A_1$, then

$$\begin{aligned} f(y) - f(x) &= (f(y) - \alpha) + (\alpha - \beta) + (\beta - f(x)) = \\ &= |f(y) - \alpha| + |\alpha - \beta| + |f(x) - \beta| \geq |f(y) - \alpha| + |\alpha - \beta| \geq 0. \end{aligned}$$

If $x \in B_2$ and $y \in A_2$, then

$$\begin{aligned} f(y) - f(x) &= (f(y) - \beta) + (\beta - f(x)) = \\ &= -|f(y) - \beta| - |f(x) - \beta| \leq -|f(y) - \beta| \leq 0. \end{aligned}$$

Thus,

$$\begin{aligned} &\langle [R_{\text{Dunkl},j}, M_f] \xi_{Q,1}, \eta_{Q,1} \rangle_{L_2(\mathbb{R}^N, w(x)dx)} \geq \\ &\geq \frac{1}{w(Q)} \int_{B_1 \times A_1} K_j(x, y) (|f(y) - \alpha| + |\alpha - \beta|) w(x) w(y) dx dy \geq \\ &\geq \frac{c_\kappa}{w^2(Q)} \int_{B_1 \times A_1} (|f(y) - \alpha| + |\alpha - \beta|) w(x) w(y) dx dy = \\ &= \frac{c_\kappa}{w^2(Q)} \cdot \left(w(B_1) \cdot \int_{A_1} |f(y) - \alpha| w(y) dy + |\alpha - \beta| w(B_1) w(A_1) \right) \geq \\ &= \frac{c_\kappa}{w^2(Q)} \cdot \left(\frac{1}{2} w(Q) \cdot \int_{A_1} |f(y) - \alpha| w(y) dy + |\alpha - \beta| \cdot \frac{1}{4} w^2(Q) \right) \geq \\ &\geq \frac{c_\kappa}{4w(Q)} \cdot \left(\int_{A_1} |f(y) - \alpha| w(y) dy + |\alpha - \beta| w(Q) \right) \end{aligned}$$

and

$$\begin{aligned} &-\langle [R_{\text{Dunkl},j}, M_f] \xi_{Q,2}, \eta_{Q,2} \rangle_{L_2(\mathbb{R}^N, w(x)dx)} \geq \\ &\geq \frac{1}{w(Q)} \int_{B_2 \times A_2} K_j(x, y) |f(y) - \beta| w(x) w(y) dx dy \geq \\ &\geq \frac{c_\kappa}{w^2(Q)} \int_{B_2 \times A_2} |f(y) - \beta| w(x) w(y) dx dy = \\ &= \frac{c_\kappa}{w^2(Q)} \cdot w(B_2) \cdot \int_{A_2} |f(y) - \beta| w(y) dy \geq \\ &= \frac{c_\kappa}{2w(Q)} \int_{A_2} |f(y) - \beta| w(y) dy. \end{aligned}$$

It remains to note that

$$\begin{aligned} &\int_{3Q} |f(y) - \alpha| w(y) dy = \int_{A_1} |f(y) - \alpha| w(y) dy + \\ &+ \int_{A_2} |f(y) - \alpha| w(y) dy + \int_{3Q \setminus (A_1 \cup A_2)} |f(y) - \alpha| w(y) dy \leq \\ &\leq \int_{A_1} |f(y) - \alpha| w(y) dy + \int_{A_2} |f(y) - \beta| w(y) dy \\ &+ \int_{A_2} |\alpha - \beta| w(y) dy + \int_{3Q \setminus (A_1 \cup A_2)} |\alpha - \beta| w(y) dy \leq \\ &\leq \int_{A_1} |f(y) - \alpha| w(y) dy + \int_{A_2} |f(y) - \beta| w(y) dy + 2|\alpha - \beta| w(3Q). \end{aligned}$$

Combining the last 3 equations, we conclude

$$\frac{c'_\kappa}{w(Q)} \int_{3Q} |f(y) - \alpha| w(y) dy \leq \max_{s=1,2} |\langle [R_{\text{Dunkl},j}, M_f] \xi_{Q,s}, \eta_{Q,s} \rangle_{L_2(\mathbb{R}^N, w(x)dx)}|.$$

If

$$|\langle [R_{\text{Dunkl},j}, M_f] \xi_{Q,1}, \eta_{Q,1} \rangle_{L_2(\mathbb{R}^N, w(x)dx)}| \geq |\langle [R_{\text{Dunkl},j}, M_f] \xi_{Q,2}, \eta_{Q,2} \rangle_{L_2(\mathbb{R}^N, w(x)dx)}|,$$

then we set $\xi_Q = \xi_{Q,1}$ and $\eta = \eta_{Q,1}$. Otherwise, set $\xi_Q = \xi_{Q,2}$ and $\eta_Q = \eta_{Q,2}$. Thus,

$$\frac{c'_\kappa}{w(Q)} \int_{3Q} |f(y) - \alpha| w(y) dy \leq |\langle [R_{\text{Dunkl},j}, M_f] \xi_Q, \eta_Q \rangle_{L_2(\mathbb{R}^N, w(x)dx)}|.$$

Hence,

$$c'_\kappa |f_{3Q} - \alpha| \leq |\langle [R_{\text{Dunkl},j}, M_f] \xi_Q, \eta_Q \rangle_{L_2(\mathbb{R}^N, w(x)dx)}|$$

and, finally,

$$\frac{c'_\kappa}{w(Q)} \int_{3Q} |f(y) - f_{3Q}| w(y) dy \leq 2 |\langle [R_{\text{Dunkl},j}, M_f] \xi_Q, \eta_Q \rangle_{L_2(\mathbb{R}^N, w(x)dx)}|.$$

□

Lemma 7.4. *Suppose that $f \in L_1^{\text{loc}}(\mathbb{R}^N)$. If $[R_{\text{Dunkl},j}, M_f] \in \mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))$ for some $j \in \{1, 2, \dots, N\}$, then*

$$\begin{aligned} & \left\| \left\{ \frac{1}{w(3Q)} \int_{3Q} |f(x) - f_{3Q}| w(x) dx \right\}_{Q \in \mathbb{D}} \right\|_{l_{N,\infty}(\mathbb{D})} \leq \\ & \leq c_\kappa \| [R_{\text{Dunkl},j}, M_f] \|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))}. \end{aligned}$$

Proof. Let $\{\xi_Q\}_{Q \in \mathbb{D}}$ and $\{\eta_Q\}_{Q \in \mathbb{D}}$ be functions constructed in Lemma 7.3. By Lemma 2.6,

$$\left\{ \langle [R_{\text{Dunkl},j}, M_f] \xi_Q, \eta_Q \rangle_{L_2(\mathbb{R}^N, w(x)dx)} \right\}_{Q \in \mathbb{D}} \in l_p.$$

The first assertion follows now from Lemma 7.3. □

Lemma 7.5. *Let Q be a cube in $\mathbb{R}^N$ (with sides parallel to coordinate hyperplanes). We have*

$$\sup_{x \in Q} w(x) \leq \frac{c_\kappa}{m(Q)} \int_Q w(x) dx.$$

Proof. If $x = c_Q + y$, $y \in [-\frac{1}{2}s(Q), \frac{1}{2}s(Q)]^d$, then

$$|\langle x, v \rangle| \leq |\langle c_Q, v \rangle| + r |\langle y, v \rangle| \leq |\langle c_Q, v \rangle| + \frac{\sqrt{d}}{2} s(Q).$$

Thus,

$$w(x) = \prod_{v \in R} |\langle x, v \rangle|^{k(v)} \leq \prod_{v \in R} (|\langle c_Q, v \rangle| + \frac{\sqrt{d}}{2} s(Q))^{k(v)}.$$

In particular,

$$LHS \leq \prod_{v \in R} (|\langle c_Q, v \rangle| + \frac{\sqrt{d}}{2} s(Q))^{k(v)}.$$

Fix chamber Γ such that $c_Q \in \Gamma$. We have

$$\int_Q w(x) dx \geq \int_{c_Q + (B(0, \frac{1}{2}s(Q)) \cap \Gamma)} w(x) dx = \int_{B(0, \frac{1}{2}s(Q)) \cap \Gamma} w(c_Q + y) dy.$$

If $y \in \Gamma$, then

$$|\langle c_Q + y, v \rangle| = |\langle c_Q, v \rangle| + |\langle y, v \rangle|, \quad v \in R.$$

Thus,

$$w(c_Q + y) = \prod_{v \in R} (|\langle c_Q, v \rangle| + |\langle y, v \rangle|)^{k(v)}.$$

Hence,

$$\int_Q w(x)dx \geq \int_{B(0, \frac{1}{2}s(Q)) \cap \Gamma} \prod_{v \in R} (|\langle c_Q, v \rangle| + |\langle y, v \rangle|)^{k(v)} dy.$$

Fix a closed cone $\Gamma' \subset \Gamma$ such that (i) interior of Γ' is non-empty and (ii) $\partial\Gamma \cap \partial\Gamma' = \{0\}$. We have

$$|\langle y, v \rangle| \geq c_{\Gamma'} |y|, \quad y \in \Gamma', \quad v \in R.$$

We, therefore, have

$$\begin{aligned} \int_Q w(x)dx &\geq \int_{B(0, \frac{1}{2}s(Q)) \cap \Gamma'} \prod_{v \in R} (|\langle c_Q, v \rangle| + c_{\Gamma'} |y|)^{k(v)} dy \geq \\ &\geq \int_{\frac{1}{4}s(Q) \leq |y| \leq \frac{1}{4}s(Q)} \prod_{v \in R} (|\langle c_Q, v \rangle| + \frac{1}{2}c_{\Gamma'} s(Q))^{k(v)} dy = \\ &= s(Q)^N \prod_{v \in R} (|\langle c_Q, v \rangle| + \frac{1}{4}c_{\Gamma'} s(Q))^{k(v)} \times (2^{-N} - 4^{-N}) \int_{\substack{y \in \Gamma' \\ |y| \leq 1}} dy. \end{aligned}$$

Combining this with the preceding paragraph, we complete the proof. $\square$

Lemma 7.6. *Suppose $1 < t < 1 + \frac{1}{|R| \cdot \max_{v \in R} \kappa(v)}$. Let Q be a cube in $\mathbb{R}^N$ (with sides parallel to coordinate hyperplanes). We have*

$$\left(\frac{\int_Q w^{1-t}(x)dx}{\int_Q w(x)dx} \right)^{\frac{1}{t}} \leq c_{t,\kappa} \frac{m(Q)}{\int_Q w(x)dx}.$$

Proof. If $v \in R$ is such that

$$\sqrt{d}s(Q) \leq |\langle c_Q, v \rangle|,$$

then

$$|\langle x, v \rangle| \geq |\langle c_Q, v \rangle| - |\langle x - c_Q, v \rangle| \geq |\langle c_Q, v \rangle| - \frac{\sqrt{d}}{2}s(Q) \geq \frac{1}{2}|\langle c_Q, v \rangle|, \quad x \in Q.$$

In this case,

$$\begin{aligned} \int_Q |\langle x, v \rangle|^{(1-t)|R|\kappa(v)} dx &\leq 2^{(t-1)|R|\kappa(v)} |\langle c_Q, v \rangle|^{(1-t)|R|\kappa(v)} \cdot \int_Q dx = \\ &= 2^{(t-1)|R|\kappa(v)} |\langle x_0, v \rangle|^{(1-t)|R|\kappa(v)} \cdot s(Q)^N. \end{aligned}$$

If $v \in R$ is such that

$$|\langle c_Q, v \rangle| \leq \sqrt{d}s(Q),$$

then let $y_0 = \langle c_Q, v \rangle$ and $z_0 = c_Q - y_0 v$. Let us introduce new variables $y = \langle x, v \rangle$ and $z = x - yv$ (it varies in the plane orthogonal to v). We write

$$\begin{aligned} \int_Q |\langle x, v \rangle|^{(1-t)|R|\kappa(v)} dx &\leq \int_{|x - c_Q| \leq \frac{\sqrt{d}}{2}s(Q)} |\langle x, v \rangle|^{(1-t)|R|\kappa(v)} dx \leq \\ &\leq \int_{|y - y_0|, |z - z_0| \leq \frac{\sqrt{d}}{2}s(Q)} |y|^{(1-t)|R|\kappa(v)} dy dz = \\ &= \left(\frac{\sqrt{d}}{2}s(Q) \right)^{N-1} \text{Vol}(\mathbb{B}^{N-1}) \cdot \int_{|y - y_0| \leq \frac{\sqrt{d}}{2}s(Q)} |y|^{(1-t)|R|\kappa(v)} dy \leq \\ &= \left(\frac{\sqrt{d}}{2}s(Q) \right)^{N-1} \text{Vol}(\mathbb{B}^{N-1}) \cdot \int_{|y| \leq \frac{3\sqrt{d}}{2}s(Q)} |y|^{(1-t)|R|\kappa(v)} dy = \end{aligned}$$

$$= s(Q)^{N+(1-t)|R|\kappa(v)} \cdot \left(\frac{\sqrt{d}}{2}\right)^{N-1} \text{Vol}(\mathbb{B}^{N-1}) \cdot \int_{|y| \leq \frac{3\sqrt{d}}{2}} |y|^{(1-t)|R|\kappa(v)} dy.$$

Combining the estimates in the preceding paragraphs, we write

$$\int_Q |\langle x, v \rangle|^{(1-t)|R|\kappa(v)} dx \leq c_1 s(Q)^N (\max\{|\langle c_Q, v \rangle|, r\})^{(1-t)|R|\kappa(v)}.$$

By Hölder inequality,

$$\begin{aligned} \int_Q w^{1-t}(x) dx &\leq \prod_{v \in R} \left(\int_Q |\langle x, v \rangle|^{(1-t)|R|\kappa(v)} dx \right)^{\frac{1}{|R|}} \leq \\ &\leq c_2 s(Q)^N \prod_{v \in R} (\max\{|\langle c_Q, v \rangle|, s(Q)\})^{(1-t)\kappa(v)}. \end{aligned}$$

The assertion follows easily. $\square$

Lemma 7.7. *Measures dx and $w(x)dx$ (and distance $(x, y) \rightarrow |x - y|_\infty$) satisfy the conditions in Proposition 1.2 in [8].*

Proof. That both measures are doubling is obvious.

Let Q be a cube in $\mathbb{R}^N$ (with sides parallel to coordinate hyperplanes). By Lemmas 7.5 and 7.6,

$$\begin{aligned} \left(\frac{1}{m(Q)} \int_Q w^t(x) dx \right)^{\frac{1}{t}} &\leq \frac{c_\kappa}{m(Q)} \int_Q w(x) dx, \quad t > 0. \\ \left(\frac{\int_Q w^{1-t}(x) dx}{\int_Q w(x) dx} \right)^{\frac{1}{t}} &\leq c_G \frac{m(Q)}{\int_Q w(x) dx}, \quad 1 < t < 1 + \frac{1}{|R| \cdot \max_{v \in R} \kappa(v)}. \end{aligned}$$

The assertion follows now from Proposition 3.6 in [8]. $\square$

Proof of Theorem 1.1 (ii). The left hand side in Lemma 7.4 is the oscillation semi-norm of the function f with respect to measure $w(x)dx$ and distance $(x, y) \rightarrow |x - y|_\infty$. So, the oscillation semi-norm of the function f with respect to measure $w(x)dx$ and distance $(x, y) \rightarrow |x - y|_\infty$ is controlled by the $\|\cdot\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))}$ -norm of the commutator $[R_{\text{Dunkl},j}, M_f]$.

Lemma 7.7 asserts that measures dx and $w(x)dx$ (and distance $(x, y) \rightarrow |x - y|_\infty$) satisfy the conditions in Proposition 1.2 in [8]. Appealing to Proposition 1.2 in [8], we conclude that the oscillation semi-norm of the function f with respect to measure dx and distance $(x, y) \rightarrow |x - y|_\infty$ is controlled by the one with respect to measure $w(x)dx$ and distance $(x, y) \rightarrow |x - y|_\infty$. By the preceding paragraph, the oscillation semi-norm of the function f with respect to measure dx and distance $(x, y) \rightarrow |x - y|_\infty$ is controlled by the $\|\cdot\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))}$ -norm of the commutator $[R_{\text{Dunkl},j}, M_f]$.

That oscillation semi-norm of the function f with respect to measure dx and distance $(x, y) \rightarrow |x - y|_\infty$ is equivalent to the Sobolev semi-norm of the function f is a standard fact. The proof (in much more general setting) is available in Proposition 1.29 in [9]. The assertion follows now from the preceding paragraph. $\square$

8. PROOF OF SMOOTHNESS

Theorem 8.1. *Let $f \in C_c^\infty(\mathbb{R}^N)$.*

(i) *for every $n \in \mathbb{N}$, $s_1, s_2 \in \mathbb{R}$ with $s_1 + s_2 = n$, the operator¹*

$$(1 + \Delta_{\text{Dunkl}})^{-\frac{s_1}{2}} [\Delta_{\text{Dunkl}}, [\Delta_{\text{Dunkl}}, [\cdots, M_f] \cdots]] (1 + \Delta_{\text{Dunkl}})^{-\frac{s_2}{2}}$$

is bounded on $L_2(\mathbb{R}^N, w(x)dx)$;

(ii) *for every $n \in \mathbb{N}$, $s_1, s_2 \in \mathbb{R}$ with $s_1 + s_2 > n$, the operator²*

$$(1 + \Delta_{\text{Dunkl}})^{-\frac{s_1}{2}} [\Delta_{\text{Dunkl}}, [\Delta_{\text{Dunkl}}, [\cdots, M_f] \cdots]] (1 + \Delta_{\text{Dunkl}})^{-\frac{s_2}{2}}$$

belongs to $\mathcal{L}_{\frac{N}{s_1+s_2-n}, \infty}(L_2(\mathbb{R}^N, w(x)dx))$;

Lemma 8.2. *Let $f \in C_c^\infty(\mathbb{R}^N)$. We have*

$$[\Delta_{\text{Dunkl}}, M_f] = 2 \sum_{j=1}^N M_{\partial_j f} T_{e_j} + \frac{1}{2} \sum_{j=1}^N \sum_{v \in R} v_j \kappa(v) M_{\delta_v f} (T_{e_j} U_v + U_v T_{e_j}) + \sum_{g \in G} M_{f_g} U_g.$$

All functions on the right hand side are smooth and compactly supported.

Proof. We have

$$\begin{aligned} [\Delta_{\text{Dunkl}}, M_f] &= \sum_{j=1}^N T_{e_j} [T_{e_j}, M_f] + [T_{e_j}, M_f] T_{e_j} = \\ &= \sum_{j=1}^N T_{e_j} M_{\partial_j f} + M_{\partial_j f} T_{e_j} + \frac{1}{2} \sum_{j=1}^N \sum_{v \in R} v_j \kappa(v) (T_{e_j} M_{\delta_v f} U_v + M_{\delta_v f} U_v T_{e_j}) = \\ &= 2 \sum_{j=1}^N M_{\partial_j f} T_{e_j} + \frac{1}{2} \sum_{j=1}^N \sum_{v \in R} v_j \kappa(v) (M_{\delta_v f} T_{e_j} U_v + M_{\delta_v f} U_v T_{e_j}) + \\ &\quad + \sum_{j=1}^N [T_{e_j}, M_{\partial_j f}] + \frac{1}{2} \sum_{j=1}^N \sum_{v \in R} v_j \kappa(v) [T_{e_j}, M_{\delta_v f}] U_v = \\ &= 2 \sum_{j=1}^N M_{\partial_j f} T_{e_j} + \frac{1}{2} \sum_{j=1}^N \sum_{v \in R} v_j \kappa(v) (M_{\delta_v f} T_{e_j} U_v + M_{\delta_v f} U_v T_{e_j}) + \\ &\quad + \sum_{j=1}^N M_{\partial_j^2 f} + \frac{1}{2} \sum_{j=1}^N \sum_{v \in R} \alpha_j \kappa(v) M_{\delta_v(\partial_j f)} U_v + \\ &\quad + \frac{1}{2} \sum_{j=1}^N \sum_{v \in R} v_j \kappa(v) M_{\partial_j(\delta_v f)} U_v + \frac{1}{4} \sum_{j=1}^N \sum_{v, v' \in R} v_j v'_j \kappa(v) \kappa(v') M_{\delta_{v'} \delta_v f} U_{v'} U_v. \end{aligned}$$

□

Lemma 8.3. *For every $f \in C_c^\infty(\mathbb{R}^N)$, the operator*

$$[\Delta_{\text{Dunkl}}, M_f] (1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}}$$

is bounded and

$$(1 + \Delta_{\text{Dunkl}})^{-1} [\Delta_{\text{Dunkl}}, M_f] \in \mathcal{L}_{N, \infty}(L_2(\mathbb{R}^N, w(x)dx)).$$

¹Commutators are repeated n times

²Commutators are repeated n times

Proof. We prove only the first assertion. The proof of the second one is similar (modulo Theorem 5.5).

Since each T_{e_j} commutes with $(1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}}$, each U_v commutes with $(1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}}$ and since the operators

$$\frac{T_{e_j}}{(1 + \Delta_{\text{Dunkl}})^{\frac{1}{2}}}, \quad 1 \leq j \leq N,$$

are bounded, the assertion follows from Lemma 8.2. $\square$

Lemma 8.4. *For every $n \in \mathbb{N}$, the operator³*

$$[\Delta_{\text{Dunkl}}, [\Delta_{\text{Dunkl}}, [\cdots, M_f] \cdots]](1 + \Delta_{\text{Dunkl}})^{-\frac{n}{2}}$$

is bounded.

Proof. Denote for brevity

$$\text{Expr}_n(f) = [\Delta_{\text{Dunkl}}, [\Delta_{\text{Dunkl}}, [\cdots, M_f] \cdots]](1 + \Delta_{\text{Dunkl}})^{-\frac{n}{2}}.$$

We prove the assertion by induction on f . The n -th statement is "for every $f \in C_c^\infty(\mathbb{R}^N)$, the operator $\text{Expr}_n(f)$ is bounded." Base of induction is given by Lemma 8.3. It suffices to prove the step of induction.

Since each T_{e_j} and each U_v commutes with Δ_{Dunkl} , it follows from Lemma 8.2 that

$$\begin{aligned} \text{Expr}_{n+1}(f) &= 2 \sum_{j=1}^N \text{Expr}_n(\partial_j f) \cdot \frac{T_{e_j}}{(1 + \Delta_{\text{Dunkl}})^{\frac{1}{2}}} + \\ &+ \frac{1}{2} \sum_{j=1}^N \sum_{v \in R} v_j \kappa(v) \text{Expr}_n(\delta_v f) \cdot \left(\frac{T_{e_j}}{(1 + \Delta_{\text{Dunkl}})^{\frac{1}{2}}} U_v + U_v \frac{T_{e_j}}{(1 + \Delta_{\text{Dunkl}})^{\frac{1}{2}}} \right) + \\ &+ \sum_{g \in G} \text{Expr}_n(f_g) \cdot (1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}} U_g. \end{aligned}$$

In each summand, the first factor is bounded by inductive assumption and the second factor is obviously bounded. This yields the step of induction and, hence, completes the proof. $\square$

Proof of Theorem 8.1 (i). The assertion follows from Lemma 8.4 and Hadamard 3 lines theorem. $\square$

Lemma 8.5. *For every $f \in C_c^\infty(\mathbb{R}^N)$, we have*

$$M_f(1 + \Delta_{\text{Dunkl}})^{-1} \in \mathcal{L}_{\frac{N}{2}, \infty}(L_2(\mathbb{R}^N, w(x)dx)).$$

Proof. Set

$$\mathcal{D} = \sum_{j=1}^N \gamma_j \otimes T_{e_j},$$

where $\{\gamma_j\}_{j=1}^N$ are Pauli matrices. We have

$$\begin{aligned} 1 \otimes M_f(1 + \Delta_{\text{Dunkl}})^{-1} &= (1 \otimes M_f)(\mathcal{D}^2 + 1)^{-1} = (1 \otimes M_f)(\mathcal{D} + i)^{-1}(\mathcal{D} - i)^{-1} = \\ &= (\mathcal{D} + i)^{-1}(1 \otimes M_f)(\mathcal{D} - i)^{-1} + [1 \otimes M_f, (\mathcal{D} + i)^{-1}](\mathcal{D} - i)^{-1} = \\ &= (\mathcal{D} + i)^{-1}(1 \otimes M_f)(\mathcal{D} - i)^{-1} + (\mathcal{D} + i)^{-1}[\mathcal{D}, 1 \otimes M_f](\mathcal{D}^2 + 1)^{-1} = \end{aligned}$$

³Commutators are repeated n times

$$\begin{aligned}
&= (\mathcal{D} + i)^{-1}(1 \otimes M_f)(\mathcal{D} - i)^{-1} + (\mathcal{D} + i)^{-1} \left(\sum_{j=1}^N \gamma_j \otimes [T_{e_j}, M_f] \right) (\mathcal{D}^2 + 1)^{-1} = \\
&= (\mathcal{D} + i)^{-1}(1 \otimes M_f)(\mathcal{D} - i)^{-1} + \sum_{j=1}^N (\mathcal{D} + i)^{-1} (\gamma_j \otimes M_{\partial_j f}) (\mathcal{D}^2 + 1)^{-1} + \\
&\quad + \frac{1}{2} \sum_{j=1}^N \sum_{v \in R} v_j \kappa(v) (\mathcal{D} + i)^{-1} (\gamma_j \otimes M_{\delta_v f} U_v) (\mathcal{D}^2 + 1)^{-1}.
\end{aligned}$$

Since Δ_{Dunkl} commutes with U_v , it follows that

$$\begin{aligned}
&1 \otimes M_f (1 + \Delta_{\text{Dunkl}})^{-1} = \\
&= (\mathcal{D} + i)^{-1}(1 \otimes M_f)(\mathcal{D} - i)^{-1} + \sum_{j=1}^N (\mathcal{D} + i)^{-1}(1 \otimes M_{\partial_j f})(\mathcal{D}^2 + 1)^{-1} \cdot (\gamma_j \otimes 1) + \\
&\quad + \frac{1}{2} \sum_{j=1}^N \sum_{v \in R} v_j \kappa(v) (\mathcal{D} + i)^{-1}(1 \otimes M_{\delta_v f})(\mathcal{D}^2 + 1)^{-1} \cdot (\gamma_j \otimes U_v).
\end{aligned}$$

Clearly,

$$\begin{aligned}
&\|(\mathcal{D} + i)^{-1}(1 \otimes M_f)(\mathcal{D} - i)^{-1}\|_{\mathcal{L}_{\frac{N}{2}, \infty}(\mathbb{C}^{2^{\lfloor \frac{N}{2} \rfloor}} \otimes L_2(\mathbb{R}^N, w(x)dx))} = \\
&= \|(\Delta_{\text{Dunkl}} + 1)^{-\frac{1}{2}} M_f (\Delta_{\text{Dunkl}} + 1)^{-\frac{1}{2}}\|_{\mathcal{L}_{\frac{N}{2}, \infty}(L_2(\mathbb{R}^N, w(x)dx))} \leq \\
&= \|\Delta_{\text{Dunkl}}^{-\frac{1}{2}} M_f \Delta_{\text{Dunkl}}^{-\frac{1}{2}}\|_{\mathcal{L}_{\frac{N}{2}, \infty}(L_2(\mathbb{R}^N, w(x)dx))} \leq \\
&\leq 2^{\frac{2}{N}} \|M_{|f|^{\frac{1}{2}}} \Delta_{\text{Dunkl}}^{-\frac{1}{2}}\|_{\mathcal{L}_{\frac{N}{2}, \infty}(L_2(\mathbb{R}^N, w(x)dx))}^2 \leq \\
&\stackrel{\text{Th. 5.5}}{\leq} c_{N, \kappa} \| |f|^{\frac{1}{2}} \|_{L_N(\mathbb{R}^N)}^2 = c_{N, \kappa} \|f\|_{L_{\frac{N}{2}}(\mathbb{R}^N)}
\end{aligned}$$

and, similarly,

$$\begin{aligned}
&\|(\mathcal{D} + i)^{-1}(1 \otimes M_{\partial_j f})(\mathcal{D}^2 + 1)^{-1}\|_{\mathcal{L}_{\frac{N}{2}, \infty}(\mathbb{C}^{2^{\lfloor \frac{N}{2} \rfloor}} \otimes L_2(\mathbb{R}^N, w(x)dx))} \leq c_{N, \kappa} \|\partial_j f\|_{L_{\frac{N}{2}}(\mathbb{R}^N)}, \\
&\|(\mathcal{D} + i)^{-1}(1 \otimes M_{\delta_v f})(\mathcal{D}^2 + 1)^{-1}\|_{\mathcal{L}_{\frac{N}{2}, \infty}(\mathbb{C}^{2^{\lfloor \frac{N}{2} \rfloor}} \otimes L_2(\mathbb{R}^N, w(x)dx))} \leq c_{N, \kappa} \|\delta_v f\|_{L_{\frac{N}{2}}(\mathbb{R}^N)}.
\end{aligned}$$

This completes the proof. $\square$

Lemma 8.6. *For every $f \in C_c^\infty(\mathbb{R}^N)$, we have*

$$M_f(1 + \Delta_{\text{Dunkl}})^{-\frac{n}{2}} \in \mathcal{L}_{\frac{N}{n}, \infty}(L_2(\mathbb{R}^N, w(x)dx)).$$

Proof. We prove the assertion by induction on f . The n -th statement is "for every $f \in C_c^\infty(\mathbb{R}^N)$, we have

$$M_f(1 + \Delta_{\text{Dunkl}})^{-\frac{n}{2}} \in \mathcal{L}_{\frac{N}{n}, \infty}(L_2(\mathbb{R}^N, w(x)dx))."$$

Base of induction (i.e. cases $n = 1$ and $n = 2$) are given by Theorem 5.5 and Lemma 8.5. It suffices to prove the step of induction.

We have

$$\begin{aligned}
&M_f(1 + \Delta_{\text{Dunkl}})^{-\frac{n+1}{2}} = \\
&= (1 + \Delta_{\text{Dunkl}})^{-1} M_f(1 + \Delta_{\text{Dunkl}})^{-\frac{n-1}{2}} + [M_f, (1 + \Delta_{\text{Dunkl}})^{-1}] (1 + \Delta_{\text{Dunkl}})^{-\frac{n-1}{2}} = \\
&= (1 + \Delta_{\text{Dunkl}})^{-1} M_f(1 + \Delta_{\text{Dunkl}})^{-\frac{n-1}{2}} + (1 + \Delta_{\text{Dunkl}})^{-1} [\Delta_{\text{Dunkl}}, M_f] (1 + \Delta_{\text{Dunkl}})^{-\frac{n+1}{2}}.
\end{aligned}$$

Choose $\phi \in C_c^\infty(\mathbb{R}^N)$ such that $(f \circ U_g) \cdot \phi = f \circ U_g$ for every $g \in G$. It follows that

$$[\Delta_{\text{Dunkl}}, M_f] = [\Delta_{\text{Dunkl}}, M_f] M_\phi.$$

Thus,

$$\begin{aligned} M_f(1 + \Delta_{\text{Dunkl}})^{-\frac{n+1}{2}} &= (1 + \Delta_{\text{Dunkl}})^{-1} M_f \cdot M_\phi(1 + \Delta_{\text{Dunkl}})^{-\frac{n-1}{2}} + \\ &+ (1 + \Delta_{\text{Dunkl}})^{-1} [\Delta_{\text{Dunkl}}, M_f] \cdot M_\phi(1 + \Delta_{\text{Dunkl}})^{-\frac{n}{2}} \cdot (1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}}. \end{aligned}$$

In the first summand on the right hand side, the first factor belongs to the ideal $\mathcal{L}_{\frac{N}{2}, \infty}(L_2(\mathbb{R}^N, w(x)dx))$ by Lemma 8.5 and the second one belongs to the ideal $\mathcal{L}_{\frac{N}{n-1}, \infty}(L_2(\mathbb{R}^N, w(x)dx))$ by inductive assumption. In the second summand on the right hand side, the first factor belongs to the ideal $\mathcal{L}_{N, \infty}(L_2(\mathbb{R}^N, w(x)dx))$ by Lemma 8.3, the second factor belongs to the ideal $\mathcal{L}_{\frac{N}{n}, \infty}(L_2(\mathbb{R}^N, w(x)dx))$ by inductive assumption and the third one is obviously bounded. $\square$

Lemma 8.7. *Let $k, l \in \mathbb{Z}_+$ with $l > k$. For every $f \in C_c^\infty(\mathbb{R}^N)$, we have*

$$T_{e_{m_1}} \cdots T_{e_{m_k}} M_f(1 + \Delta_{\text{Dunkl}})^{-\frac{l}{2}}$$

belongs to $\mathcal{L}_{\frac{N}{l-k}, \infty}(L_2(\mathbb{R}^N, w(x)dx))$.

Proof. We prove the assertion by induction on k . Base of induction (i.e., the case $k = 0$) is established in Lemma 8.6.

We have

$$\begin{aligned} T_{e_{m_1}} \cdots T_{e_{m_k}} M_f(1 + \Delta_{\text{Dunkl}})^{-\frac{l}{2}} &= T_{e_{m_1}} \cdots T_{e_{m_{k-1}}} M_f T_{e_{m_k}}(1 + \Delta_{\text{Dunkl}})^{-\frac{l}{2}} + \\ &+ T_{e_{m_1}} \cdots T_{e_{m_{k-1}}} [T_{e_{m_k}}, M_f](1 + \Delta_{\text{Dunkl}})^{-\frac{l}{2}} = \\ &= T_{e_{m_1}} \cdots T_{e_{m_{k-1}}} M_f(1 + \Delta_{\text{Dunkl}})^{-\frac{l-1}{2}} \times \frac{T_{e_{m_k}}}{(1 + \Delta_{\text{Dunkl}})^{\frac{1}{2}}} + \\ &+ T_{e_{m_1}} \cdots T_{e_{m_{k-1}}} M_{\partial_{m_k}} f(1 + \Delta_{\text{Dunkl}})^{-\frac{l-1}{2}} \times (1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}} + \\ &+ \frac{1}{2} \sum_{v \in R} v_{m_k} \kappa(v) T_{e_{m_1}} \cdots T_{e_{m_{k-1}}} M_{\delta_v} f(1 + \Delta_{\text{Dunkl}})^{-\frac{l-1}{2}} \times (1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}} U_v. \end{aligned}$$

In each summand on the right hand side, the first factor belongs to the ideal $\mathcal{L}_{\frac{N}{l-k}, \infty}(L_2(\mathbb{R}^N, w(x)dx))$ by inductive assumption and the second factor is obviously bounded. This yields the step of induction and, hence, completes the proof. $\square$

Lemma 8.8. *Let $s_1, s_2 \in \mathbb{R}$ be such that $s_1 + s_2 > 0$. We have*

$$(1 + \Delta_{\text{Dunkl}})^{-\frac{s_1}{2}} M_f(1 + \Delta_{\text{Dunkl}})^{-\frac{s_2}{2}}$$

belongs to $\mathcal{L}_{\frac{N}{s_1+s_2}, \infty}(L_2(\mathbb{R}^N, w(x)dx))$.

Proof. Step 1: Let us prove the assertion for $s_1, s_2 \in \mathbb{Z}$.

Suppose first $s_1 \leq 0$. Denote $k = -s_1 \geq 0$ and $l = s_2 > k$. Denote for brevity

$$V = (1 + \Delta_{\text{Dunkl}})^{\frac{k}{2}} \left(1 + \sum_{j_1, \dots, j_k=1}^N |T_{e_{j_1}} \cdots T_{e_{j_k}}|\right)^{-1}.$$

Obviously, V is bounded. We have

$$\begin{aligned} (1 + \Delta_{\text{Dunkl}})^{\frac{k}{2}} &= V \times (1 + \sum_{m_1, \dots, m_k=1}^N |T_{e_{m_1}} \cdots T_{e_{m_k}}|) = V + \\ &+ \sum_{m_1, \dots, m_k=1}^N V \times \text{sgn}(T_{e_{m_1}} \cdots T_{e_{m_k}})^* \times T_{e_{m_1}} \cdots T_{e_{m_k}}. \end{aligned}$$

It, therefore, suffices to prove that operators

$$M_f(1 + \Delta_{\text{Dunkl}})^{-\frac{l}{2}}, \quad T_{e_{m_1}} \cdots T_{e_{m_k}} M_f(1 + \Delta_{\text{Dunkl}})^{-\frac{l}{2}}$$

belong to $\mathcal{L}_{\frac{N}{s_1+s_2}, \infty}(L_2(\mathbb{R}^N, w(x)dx))$. This is established in Lemma 8.6 and Lemma 8.7, respectively. Hence, the claim is proved for $s_1, s_2 \in \mathbb{Z}$ and $s_1 \leq 0$.

If $s_2 \leq 0$, then the claim follows from that for $s_2 \leq 0$ by taking the adjoint. If $s_1, s_2 \geq 0$, then, according to Lemma 8.7 the operators

$$(1 + \Delta_{\text{Dunkl}})^{-\frac{s_1+s_2}{2}} M_f, \quad M_f(1 + \Delta_{\text{Dunkl}})^{-\frac{s_1+s_2}{2}}$$

belong to $\mathcal{L}_{\frac{N}{s_1+s_2}, \infty}(L_2(\mathbb{R}^N, w(x)dx))$. The claim for $s_1, s_2 \geq 0$ follows now from the Hadamard 3 lines theorem.

Step 2: The general case follows now from Step 1 by complex interpolation. $\square$

Proof of Theorem 8.1 (ii). Choose $\phi \in C_c^\infty(\mathbb{R}^N)$ such that $(f \circ U_g) \cdot \phi = f \circ U_g$ for every $g \in G$. It follows that

$$[\Delta_{\text{Dunkl}}, M_f] = [\Delta_{\text{Dunkl}}, M_f] M_\phi.$$

We have

$$\begin{aligned} &(1 + \Delta_{\text{Dunkl}})^{-\frac{s_1}{2}} [\Delta_{\text{Dunkl}}, [\Delta_{\text{Dunkl}}, [\cdots, M_f] \cdots]] (1 + \Delta_{\text{Dunkl}})^{-\frac{s_2}{2}} = \\ &= (1 + \Delta_{\text{Dunkl}})^{-\frac{s_1}{2}} [\Delta_{\text{Dunkl}}, [\Delta_{\text{Dunkl}}, [\cdots, M_f] \cdots]] M_\phi (1 + \Delta_{\text{Dunkl}})^{-\frac{s_2}{2}} = \\ &= (1 + \Delta_{\text{Dunkl}})^{-\frac{s_1}{2}} [\Delta_{\text{Dunkl}}, [\Delta_{\text{Dunkl}}, [\cdots, M_f] \cdots]] (1 + \Delta_{\text{Dunkl}})^{-\frac{n-s_1}{2}} \times \\ &\quad \times (1 + \Delta_{\text{Dunkl}})^{-\frac{s_1-n}{2}} M_\phi (1 + \Delta_{\text{Dunkl}})^{-\frac{s_2}{2}}. \end{aligned}$$

The first factor is bounded by Theorem 8.1 (i). The second factor belongs to $\mathcal{L}_{\frac{N}{s_1+s_2-n}, \infty}(L_2(\mathbb{R}^N, w(x)dx))$ by Lemma 8.8. $\square$

9. APPROXIMATION THEOREM

Theorem 9.1. *For every $f \in C_c^\infty(\mathbb{R}^N)$, we have*

$$\begin{aligned} M_{w^{\frac{1}{2}}} [R_{\text{Dunkl}, j}, M_f] M_{w^{-\frac{1}{2}}} &\in V \Delta^{-\frac{1}{2}} + (\mathcal{L}_{N, \infty})_0(L_2(\mathbb{R}^N)), \\ V &= M_{\partial_j f} + \frac{1}{2} \sum_{v \in R} v_k \kappa(v) M_{\delta_v f} U_v - \sum_{k=1}^N R_j M_{\partial_k f} R_k - \\ &\quad - \frac{1}{4} R_j \cdot \sum_{k=1}^N \sum_{v \in R} v_k \kappa(v) M_{\delta_v f} (R_k U_v + U_v R_k). \end{aligned}$$

9.1. First approximation theorem.

Theorem 9.2. *For every $f \in C_c^\infty(\mathbb{R}^N)$, we have*

$$\begin{aligned} & [(1 + \Delta_{\text{Dunkl}})^{\frac{1}{2}}, M_f](1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}} - \frac{1}{2}[\Delta_{\text{Dunkl}}, M_f](1 + \Delta_{\text{Dunkl}})^{-1} \in \\ & \in \mathcal{L}_{\frac{2N}{3}, \infty}(L_2(\mathbb{R}^N, w(x)dx)). \end{aligned}$$

Lemma 9.3. *Let $B \geq 1$. Suppose*

$$B^{-1}[B^2, [B^2, A]]B^{-\frac{5}{2}} \in \mathcal{L}_{\frac{2N}{3}, \infty}.$$

We have

$$[B, A]B^{-1} - \frac{1}{2}[B^2, A]B^{-2} \in \mathcal{L}_{\frac{2N}{3}, \infty}.$$

Proof. We have

$$B = \frac{2}{\pi} \int_0^\infty \frac{B^2 d\lambda}{\lambda^2 + B^2}.$$

Hence,

$$[B, A]B^{-1} = \frac{2}{\pi} \int_0^\infty \left[\frac{B^2}{\lambda^2 + B^2}, A \right] B^{-1} d\lambda.$$

Obviously,

$$\begin{aligned} \left[\frac{B^2}{\lambda^2 + B^2}, A \right] B^{-1} &= - \left[\frac{\lambda^2}{\lambda^2 + B^2}, A \right] B^{-1} = \frac{\lambda^2}{\lambda^2 + B^2} [B^2, A] \frac{1}{B(\lambda^2 + B^2)} = \\ &= [B^2, A] \frac{\lambda^2}{B(\lambda^2 + B^2)^2} - \frac{\lambda^2}{\lambda^2 + B^2} [B^2, [B^2, A]] \frac{1}{B(\lambda^2 + B^2)^2}. \end{aligned}$$

Obviously,

$$\begin{aligned} \int_0^\infty [B^2, A] \frac{\lambda^2}{B(\lambda^2 + B^2)^2} d\lambda &= [B^2, A] B^{-1} \cdot \int_0^\infty \frac{\lambda^2}{(\lambda^2 + B^2)^2} d\lambda = \\ &= [B^2, A] B^{-2} \cdot \int_0^\infty \frac{\lambda^2}{(\lambda^2 + 1)^2} d\lambda = \frac{\pi}{4} [B^2, A] B^{-2}. \end{aligned}$$

On the other hand,

$$\begin{aligned} & \left\| \frac{\lambda^2}{\lambda^2 + B^2} [B^2, [B^2, A]] \frac{1}{B(\lambda^2 + B^2)^2} \right\|_{\frac{2N}{3}, \infty} \leq \\ & \leq \left\| \frac{\lambda^2 B}{\lambda^2 + B^2} \right\|_\infty \|B^{-1}[B^2, [B^2, A]]B^{-\frac{5}{2}}\|_{\frac{2N}{3}, \infty} \left\| \frac{B^{\frac{3}{2}}}{(\lambda^2 + B^2)^2} \right\|_\infty. \end{aligned}$$

Next,

$$\left\| \frac{\lambda^2 B}{\lambda^2 + B^2} \right\|_\infty \leq \sup_{t \geq 1} \frac{\lambda^2 t}{\lambda^2 + t^2} \leq \lambda$$

and

$$\left\| \frac{B^{\frac{3}{2}}}{(\lambda^2 + B^2)^2} \right\|_\infty \leq \sup_{t \geq 1} \frac{t^{\frac{3}{2}}}{(\lambda^2 + t^2)^2} \leq (\lambda^2 + 1)^{-\frac{5}{4}}.$$

Thus,

$$\begin{aligned} & \left\| \frac{\lambda^2}{\lambda^2 + B^2} [B^2, [B^2, A]] \frac{1}{B(\lambda^2 + B^2)^2} \right\|_{\frac{2N}{3}, \infty} \leq \\ & \leq \frac{\lambda}{(\lambda^2 + 1)^{\frac{5}{4}}} \|B^{-1}[B^2, [B^2, A]]B^{-\frac{5}{2}}\|_{\frac{2N}{3}, \infty}. \end{aligned}$$

Hence, the integral

$$\int_0^\infty \frac{\lambda^2}{\lambda^2 + B^2} [B^2, [B^2, A]] \frac{1}{B(\lambda^2 + B^2)^2} d\lambda$$

converges absolutely in $\mathcal{L}_{\frac{2N}{3}, \infty}$ and the assertion follows. $\square$

Proof of Theorem 9.2. The assertion follows from Theorem 8.1 and Lemma 9.3. $\square$

9.2. Second approximation theorem.

Theorem 9.4. *If $f \in C_c^\infty(\mathbb{R}^N)$, then*

$$M_f(1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}} - M_f(1 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}})^{-\frac{1}{2}} \in (\mathcal{L}_{N, \infty})_0(L_2(\mathbb{R}^N, w(x)dx)).$$

Lemma 9.5. *Let A, B_1, B_2 be such that $B_1, B_2 \geq 1$. Suppose that*

- (i) $A(B_2 - B_1)B_2^{-\frac{1}{2}}, [B_1, A]B_1^{-\frac{1}{2}}$ and $[B_1, A]B_2^{-\frac{1}{2}}$ are bounded;
- (ii) *there exists A' with $B_1^{-\frac{3}{4}}A' \in \mathcal{L}_{\frac{2N}{3}, \infty}$ such that $A = A'A$ and $[B_1, A] = A'[B_1, A]$;*

We have

$$AB_1^{-\frac{1}{2}} - AB_2^{-\frac{1}{2}} \in \mathcal{L}_{\frac{2N}{3}, \infty}.$$

Proof. We have

$$B_k^{-\frac{1}{2}} = \frac{2}{\pi} \int_0^\infty \frac{d\lambda}{\lambda^2 + B_k}, \quad k = 1, 2.$$

Thus,

$$AB_1^{-\frac{1}{2}} - AB_2^{-\frac{1}{2}} = \int_0^\infty A \left(\frac{1}{\lambda^2 + B_1} - \frac{1}{\lambda^2 + B_2} \right) d\lambda.$$

Obviously,

$$\begin{aligned} A \left(\frac{1}{\lambda^2 + B_1} - \frac{1}{\lambda^2 + B_2} \right) &= A \frac{1}{\lambda^2 + B_1} (B_2 - B_1) \frac{1}{\lambda^2 + B_2} = \\ &= \frac{1}{\lambda^2 + B_1} A(B_2 - B_1) \frac{1}{\lambda^2 + B_2} + \frac{1}{\lambda^2 + B_1} [B_1, A] \frac{1}{\lambda^2 + B_1} (B_2 - B_1) \frac{1}{\lambda^2 + B_2} = \\ &= \frac{1}{\lambda^2 + B_1} A(B_2 - B_1) \frac{1}{\lambda^2 + B_2} + \frac{1}{\lambda^2 + B_1} [B_1, A] \frac{1}{\lambda^2 + B_1} - \frac{1}{\lambda^2 + B_1} [B_1, A] \frac{1}{\lambda^2 + B_2}. \end{aligned}$$

Consider, for example, the first summand. We write

$$\begin{aligned} \frac{1}{\lambda^2 + B_1} A(B_2 - B_1) \frac{1}{\lambda^2 + B_2} &= \frac{1}{\lambda^2 + B_1} A' \cdot A(B_2 - B_1) \frac{1}{\lambda^2 + B_2} = \\ &= \frac{B_1^{\frac{3}{4}}}{\lambda^2 + B_1} \cdot B_1^{-\frac{3}{4}} A' \cdot A(B_2 - B_1) B_2^{-\frac{1}{2}} \cdot \frac{B_2^{\frac{1}{2}}}{\lambda^2 + B_2}. \end{aligned}$$

Thus,

$$\begin{aligned} &\left\| \frac{1}{\lambda^2 + B_1} A(B_2 - B_1) \frac{1}{\lambda^2 + B_2} \right\|_{\frac{3N}{4}, \infty} \leq \\ &\leq \left\| \frac{B_1^{\frac{3}{4}}}{\lambda^2 + B_1} \right\|_\infty \cdot \left\| B_1^{-\frac{3}{4}} A' \right\|_{\frac{3N}{4}, \infty} \cdot \|A(B_2 - B_1) B_2^{-\frac{1}{2}}\|_\infty \cdot \left\| \frac{B_2^{\frac{1}{2}}}{\lambda^2 + B_2} \right\|_\infty \leq \\ &\leq \sup_{t \geq 1} \frac{t^{\frac{3}{4}}}{\lambda^2 + t} \cdot \left\| B_1^{-\frac{3}{4}} A' \right\|_{\frac{3N}{4}, \infty} \cdot \|A(B_2 - B_1) B_2^{-\frac{1}{2}}\|_\infty \cdot \sup_{t \geq 1} \frac{t^{\frac{1}{2}}}{\lambda^2 + t} \leq \\ &\leq (\lambda^2 + 1)^{-\frac{1}{4}} \cdot \left\| B_1^{-\frac{3}{4}} A' \right\|_{\frac{3N}{4}, \infty} \cdot \|A(B_2 - B_1) B_2^{-\frac{1}{2}}\|_\infty \cdot (\lambda^2 + 1)^{-\frac{1}{2}}. \end{aligned}$$

Hence, the first summand is absolutely integrable in $\mathcal{L}_{\frac{3N}{4}, \infty}$. Similarly, so are the other summands. This completes the proof. $\square$

Lemma 9.6. *Let P be a differential operator of degree 1 with smooth compactly supported coefficients. Suppose that the coefficients are supported away from the hyperplanes $\{\{x : \langle x, v \rangle = 0\}\}_{v \in R}$. The operators*

$$P(1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}}, \quad P(1 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}})^{-\frac{1}{2}}$$

are bounded on $L_2(\mathbb{R}^N, w(x)dx)$.

Proof. By assumption, $P = M_{w^{-\frac{1}{2}}} P' M_{w^{\frac{1}{2}}}$, where P' is a differential operator of degree 1 with smooth compactly supported coefficients. Thus,

$$P(1 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}})^{-\frac{1}{2}} = M_{w^{-\frac{1}{2}}} \cdot P'(1 + \Delta)^{-\frac{1}{2}} \cdot M_{w^{\frac{1}{2}}}.$$

Boundedness of the right hand side is obvious.

Similarly,

$$P = \sum_{j=1}^N M_{f_j} T_{e_j} + \sum_{g \in G} M_{h_g} U_g,$$

where functions $\{f_j\}_{j=1}^N$ and $\{h_g\}_{g \in G}$ are smooth and compactly supported. Since the operators $\{U_g\}_{g \in G}$ commute with Δ_{Dunkl} , it follows that

$$P(1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}} = \sum_{j=1}^N M_{f_j} \frac{T_{e_j}}{(1 + \Delta_{\text{Dunkl}})^{\frac{1}{2}}} + \sum_{g \in G} M_{h_g} \frac{T_{e_j}}{(1 + \Delta_{\text{Dunkl}})^{\frac{1}{2}}} U_g.$$

Boundedness of the right hand side is obvious. $\square$

Proof of Theorem 9.4. Step 1: Suppose first that f is supported away from the hyperplanes $\{\{x : \langle x, v \rangle = 0\}\}_{v \in R}$.

We apply Lemma 9.5 with $A = M_f$, $B_1 = 1 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}}$ and $B_2 = 1 + \Delta_{\text{Dunkl}}$.

Choose $\phi \in C_c^\infty(\mathbb{R}^N)$ such that $f = f\phi$ and set $A' = M_\phi$. We have

$$M_\phi[M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}}, M_f] = [M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}}, M_f].$$

Thus, $A = A'A$ and $[B_1, A] = A'[B_1, A]$. It follows from the usual Cwikel estimates that $B_1^{-\frac{3}{4}} A' \in \mathcal{L}_{\frac{2N}{3}, \infty}(L_2(\mathbb{R}^N, w(x)dx))$. This verifies the assumption (ii) in Lemma 9.5.

Note that $[B_1, A]$ is a differential operator P of degree 1 with smooth compactly supported coefficients. Furthermore, the coefficients are supported away from the hyperplanes $\{\{x : \langle x, v \rangle = 0\}\}_{v \in R}$. By Lemma 9.6, the operators

$$[B_1, A]B_1^{-\frac{1}{2}} = P(1 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}})^{-\frac{1}{2}}, \quad [B_1, A]B_2^{-\frac{1}{2}} = P(1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}},$$

are bounded on $L_2(\mathbb{R}^N, w(x)dx)$. Note that there exist differential operators $\{P_g\}_{g \in G}$ of degree 1 with smooth compactly supported coefficients such that the coefficients are supported away from the hyperplanes $\{\{x : \langle x, v \rangle = 0\}\}_{v \in R}$ and such that

$$A(B_2 - B_1) = \sum_{g \in G} P_g U_g.$$

Since $\{U_g\}_{g \in G}$ commute with Δ_{Dunkl} , it follows that

$$A(B_2 - B_1) = \sum_{g \in G} P_g (1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}} \cdot U_g$$

is bounded on $L_2(\mathbb{R}^N, w(x)dx)$. This verifies the assumption (i) in Lemma 9.5.

Using Lemma 9.5, we infer the claim of Step 1.

Step 2: Choose a sequence $\{\phi_n\}_{n \geq 1} \subset C_c^\infty(\mathbb{R}^N)$ such that (a) $0 \leq \phi_n \leq 1$ for $n \geq 1$ (b) ϕ_n is supported away from the hyperplanes $\{x : \langle x, v \rangle = 0\}$, $v \in R$, for $n \geq 1$ (c) $\text{supp}(\phi_n) \uparrow \mathbb{R}^N$ as $n \rightarrow \infty$.

By Step 1,

$$M_{f\phi_n}(1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}} - M_{f\phi_n}(1 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}})^{-\frac{1}{2}} \in (\mathcal{L}_{N,\infty})_0(L_2(\mathbb{R}^N, w(x)dx))$$

for every $n \geq 1$.

By Theorem 5.5,

$$M_{f\phi_n}(1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}} \rightarrow M_f(1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}}$$

in $\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))$ as $n \rightarrow \infty$. By the usual Cwikel estimate,

$$M_{f\phi_n}(1 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}})^{-\frac{1}{2}} \rightarrow M_f(1 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}})^{-\frac{1}{2}}$$

in $\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))$ as $n \rightarrow \infty$. Thus

$$\begin{aligned} \lim_{n \rightarrow \infty} M_{f\phi_n}(1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}} - M_{f\phi_n}(1 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}})^{-\frac{1}{2}} &= \\ &= M_f(1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}} - M_f(1 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}})^{-\frac{1}{2}}, \end{aligned}$$

where the limit is understood in $\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))$. Since the separable part $(\mathcal{L}_{N,\infty})_0(L_2(\mathbb{R}^N, w(x)dx))$ is closed in $\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))$, the assertion follows. $\square$

9.3. Proof of Theorem 9.1.

Lemma 9.7. *We have*

$$[R_{\text{Dunkl},j}, M_f] - [\frac{T_{e_j}}{(1 + \Delta_{\text{Dunkl}})^{\frac{1}{2}}}, M_f] \in \mathcal{L}_{\frac{N}{2},\infty}(L_2(\mathbb{R}^N, w(x)dx)).$$

Proof. Set

$$h(x) = 1 - \frac{|x|}{(1 + |x|^2)^{\frac{1}{2}}}, \quad x \in \mathbb{R}^N.$$

We, obviously, have

$$R_{\text{Dunkl},j} - \frac{T_{e_j}}{(1 + \Delta_{\text{Dunkl}})^{\frac{1}{2}}} = R_{\text{Dunkl},j} \cdot h(\nabla_{\text{Dunkl}}).$$

Thus,

$$\begin{aligned} [R_{\text{Dunkl},j}, M_f] - [\frac{T_{e_j}}{(1 + \Delta_{\text{Dunkl}})^{\frac{1}{2}}}, M_f] &= \\ &= R_{\text{Dunkl},j} \cdot h(\nabla_{\text{Dunkl}}) M_f - M_f h(\nabla_{\text{Dunkl}}) \cdot R_{\text{Dunkl},j} = \\ &= R_{\text{Dunkl},j} \cdot h(\nabla_{\text{Dunkl}}) (1 + \Delta_{\text{Dunkl}}) \cdot (1 + \Delta_{\text{Dunkl}})^{-1} M_f - \\ &\quad - M_f (1 + \Delta_{\text{Dunkl}})^{-1} \cdot (1 + \Delta_{\text{Dunkl}}) h(\nabla_{\text{Dunkl}}) \cdot R_{\text{Dunkl},j}. \end{aligned}$$

In the first summand on the right hand side, the first and the second factors are bounded and the third one belongs to $\mathcal{L}_{\frac{N}{2},\infty}(L_2(\mathbb{R}^N, w(x)dx))$ by Lemma 8.6. In the second summand on the right hand side, the second and the third factors are bounded and the first one belongs to $\mathcal{L}_{\frac{N}{2},\infty}(L_2(\mathbb{R}^N, w(x)dx))$ by Lemma 8.6. $\square$

Lemma 9.8. *Let $\{B_j\}_{j=1}^N$ and $\{D_j\}_{j=1}^N$ be commuting tuples of self-adjoint operators. Denote $\langle B \rangle = (1 + \sum_{j=1}^N B_j^2)^{\frac{1}{2}}$ and $\langle D \rangle = (1 + \sum_{j=1}^N D_j^2)^{\frac{1}{2}}$. Suppose that there exists a sequence $\{\phi_n\}_{n \geq 1} \subset C_c^\infty(\mathbb{R}^N)$ with*

- (i) $0 \leq \phi_n \leq 1$ for $n \geq 1$ and $\text{supp}(\phi_n) \uparrow \mathbb{R}^N$ as $n \rightarrow \infty$;
- (ii)

$$M_{\phi_n} \langle B \rangle^{-1} - M_{\phi_n} \langle D \rangle^{-1} \in (\mathcal{L}_{N,\infty})_0(L_2(\mathbb{R}^N, w(x)dx)), \quad n \geq 1;$$

- (iii) $M_{\phi_n}(B_k - D_k)$ is bounded⁴ and compactly supported for $n \geq 1$ and $1 \leq k \leq K$;
- (iv) $[D_k, M_{f\phi_n}]$ is bounded⁵ and compactly supported for $n \geq 1$ and $1 \leq k \leq K$;
- (v) $[\langle D \rangle^{-1}, M_{f\phi_n}] \in (\mathcal{L}_{N,\infty})_0(\mathbb{R}^N, w(x)dx)$ for every $n \geq 1$;
- (vi)

$$\|M_h \langle B \rangle^{-1}\|_{\mathcal{L}_{N,\infty}(\mathbb{R}^N, w(x)dx)} \leq c_N \|h\|_{L_N(\mathbb{R}^N)}, \quad h \in L_N(\mathbb{R}^N),$$

$$\|M_h \langle D \rangle^{-1}\|_{\mathcal{L}_{N,\infty}(\mathbb{R}^N, w(x)dx)} \leq c_N \|h\|_{L_N(\mathbb{R}^N)}, \quad h \in L_N(\mathbb{R}^N);$$

We have

$$\frac{B_j}{\langle B \rangle} M_f \frac{B_k}{\langle B \rangle^2} - \frac{D_j}{\langle D \rangle} M_f \frac{D_k}{\langle D \rangle^2} \in (\mathcal{L}_{N,\infty})_0(L_2(\mathbb{R}^N, w(x)dx)).$$

Proof. Choose real-valued $\psi_n \in C_c^\infty(\mathbb{R}^N)$ such that

$$M_{\phi_n}(B_k - D_k) = M_{\phi_n}(B_k - D_k) \cdot M_{\psi_n}, \quad 1 \leq k \leq K,$$

$$[M_{f\phi_n}, D_k] = M_{\psi_n} \cdot [M_{f\phi_n}, D_k] \cdot M_{\psi_n}, \quad 1 \leq k \leq K.$$

We have

$$\begin{aligned} & \frac{B_j}{\langle B \rangle} M_{f\phi_n} \frac{B_k}{\langle B \rangle^2} - \frac{D_j}{\langle D \rangle} M_{f\phi_n} \frac{D_k}{\langle D \rangle^2} = \frac{B_j}{\langle B \rangle} \cdot M_f (M_{\phi_n} \langle B \rangle^{-1} - M_{\phi_n} \langle D \rangle^{-1}) \cdot \frac{B_k}{\langle B \rangle} + \\ & + \frac{B_j}{\langle B \rangle} \cdot [M_{f\phi_n}, \langle D \rangle^{-1}] \cdot \frac{B_k}{\langle B \rangle} + \frac{B_j}{\langle B \rangle} \cdot \langle D \rangle^{-1} M_f \cdot M_{\phi_n}(B_k - D_k) \cdot M_{\psi_n} \langle B \rangle^{-1} + \\ & + \frac{B_j}{\langle B \rangle} \cdot \langle D \rangle^{-1} M_{\psi_n} \cdot [M_{f\phi_n}, D_k] \cdot M_{\psi_n} \langle B \rangle^{-1} + \frac{B_j}{\langle B \rangle} \frac{D_k}{\langle D \rangle} \cdot M_f (M_{\phi_n} \langle B \rangle^{-1} - M_{\phi_n} \langle D \rangle^{-1}) + \\ & + \frac{B_j}{\langle B \rangle} \cdot \langle D \rangle^{-1} M_{\psi_n} \cdot [D_k, M_{f\phi_n}] \cdot M_{\psi_n} \langle D \rangle^{-1} + \frac{B_j}{\langle B \rangle} \cdot [\langle D \rangle^{-1}, M_{f\phi_n}] \cdot \frac{D_k}{\langle D \rangle} + \\ & + \langle B \rangle^{-1} M_{\psi_n} \cdot (B_j - D_j) M_{\phi_n} \cdot M_f \frac{D_k}{\langle D \rangle^2} + \\ & + \langle B \rangle^{-1} M_{\psi_n} \cdot [D_j, M_{f\phi_n}] \cdot M_{\psi_n} \frac{D_k}{\langle D \rangle^2} + (\langle B \rangle^{-1} - \langle D \rangle^{-1}) M_{\phi_n} \cdot M_f \frac{D_j D_k}{\langle D \rangle^2} + \\ & + \langle D \rangle^{-1} M_{\psi_n} \cdot [M_{f\phi_n}, D_j] \cdot M_{\psi_n} \frac{D_k}{\langle D \rangle^2}. \end{aligned}$$

It follows now from the assumptions and from Hölder inequality that

$$\frac{B_j}{\langle B \rangle} M_{f\phi_n} \frac{B_k}{\langle B \rangle^2} - \frac{D_j}{\langle D \rangle} M_{f\phi_n} \frac{D_k}{\langle D \rangle^2} \in (\mathcal{L}_{N,\infty})_0(L_2(\mathbb{R}^N, w(x)dx)).$$

Obviously,

$$\left\| \frac{B_j}{\langle B \rangle} M_{f\phi_n} \frac{B_k}{\langle B \rangle^2} - \frac{D_j}{\langle D \rangle} M_{f\phi_n} \frac{D_k}{\langle D \rangle^2} \right\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))} \leq$$

⁴By a constant which depends also on n .

⁵By a constant which depends also on n .

$$\begin{aligned}
&\leq \left\| M_{f(1-\phi_n)} \langle B \rangle^{-1} \right\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))} \leq c_N \|f(1-\phi_n)\|_{L_N(\mathbb{R}^N)}, \\
&\quad \left\| \frac{D_j}{\langle D \rangle} M_{f\phi_n} \frac{D_k}{\langle D \rangle^2} - \frac{D_j}{\langle D \rangle} M_f \frac{D_k}{\langle D \rangle^2} \right\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))} \leq \\
&\leq \left\| M_{f(1-\phi_n)} \langle D \rangle^{-1} \right\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))} \leq c_N \|f(1-\phi_n)\|_{L_N(\mathbb{R}^N)}.
\end{aligned}$$

Thus,

$$\begin{aligned}
&\lim_{n \rightarrow \infty} \frac{B_j}{\langle B \rangle} M_{f\phi_n} \frac{B_k}{\langle B \rangle^2} - \frac{D_j}{\langle D \rangle} M_{f\phi_n} \frac{D_k}{\langle D \rangle^2} = \\
&= \frac{B_j}{\langle B \rangle} M_f \frac{B_k}{\langle B \rangle^2} - \frac{D_j}{\langle D \rangle} M_f \frac{D_k}{\langle D \rangle^2},
\end{aligned}$$

where the limit is take in $\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))$. Since $(\mathcal{L}_{N,\infty})_0(L_2(\mathbb{R}^N, w(x)dx))$ is closed in $\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))$, the assertion follows. $\square$

Lemma 9.9. For every $f \in C_c^\infty(\mathbb{R}^N)$,

$$\begin{aligned}
&\frac{T_{e_j}}{(1 + \Delta_{\text{Dunkl}})^{\frac{1}{2}}} M_f \frac{T_{e_k}}{1 + \Delta_{\text{Dunkl}}} U_v \in \\
&\in M_{w^{-\frac{1}{2}}} R_j M_f R_k U_v \Delta^{-\frac{1}{2}} M_{w^{\frac{1}{2}}} + (\mathcal{L}_{N,\infty})_0(L_2(\mathbb{R}^N, w(x)dx)), \\
&\frac{T_{e_j}}{(1 + \Delta_{\text{Dunkl}})^{\frac{1}{2}}} M_f U_v \frac{T_{e_j}}{1 + \Delta_{\text{Dunkl}}} \in \\
&\in M_{w^{-\frac{1}{2}}} R_j M_f U_v R_k \Delta^{-\frac{1}{2}} M_{w^{\frac{1}{2}}} + (\mathcal{L}_{N,\infty})_0(L_2(\mathbb{R}^N, w(x)dx)).
\end{aligned}$$

Proof. The assertion follows by applying Lemma 9.8 to the tuples $(T_{e_1}, \dots, T_{e_N})$ and $(M_{w^{-\frac{1}{2}}} \partial_1 M_{w^{\frac{1}{2}}}, \dots, M_{w^{-\frac{1}{2}}} \partial_N M_{w^{\frac{1}{2}}})$ and sequence $\{\phi_n\}_{n \geq 1} \subset C_c^\infty(\mathbb{R}^N)$ supported away from the hyperplanes $\{x : \langle x, v \rangle = 0\}_{v \in R}$. $\square$

Proof of Theorem 9.1. We have

$$\begin{aligned}
&[\frac{T_{e_j}}{(1 + \Delta_{\text{Dunkl}})^{\frac{1}{2}}}, M_f] = \\
&= [T_{e_j}, M_f](1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}} - \frac{T_{e_j}}{(1 + \Delta_{\text{Dunkl}})^{\frac{1}{2}}} \cdot [(1 + \Delta_{\text{Dunkl}})^{\frac{1}{2}}, M_f](1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}}.
\end{aligned}$$

Using Lemma 9.7 and Theorem 9.2, we write

$$\begin{aligned}
&[R_{\text{Dunkl},j}, M_f] \in [T_{e_j}, M_f](1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}} - \\
&-\frac{1}{2} \cdot \frac{T_{e_j}}{(1 + \Delta_{\text{Dunkl}})^{\frac{1}{2}}} \cdot [\Delta_{\text{Dunkl}}, M_f](1 + \Delta_{\text{Dunkl}})^{-1} \in (\mathcal{L}_{N,\infty})_0(L_2(\mathbb{R}^N, w(x)dx)).
\end{aligned}$$

It follows from Lemma 8.2 and Lemma 8.6 that

$$\begin{aligned}
&[\Delta_{\text{Dunkl}}, M_f](1 + \Delta_{\text{Dunkl}})^{-1} - 2 \sum_{k=1}^N M_{\partial_k f} \frac{T_{e_k}}{1 + \Delta_{\text{Dunkl}}} - \\
&-\frac{1}{2} \sum_{k=1}^N \sum_{v \in R} v_k \kappa(v) M_{\delta_v f} \left(\frac{T_{e_k}}{1 + \Delta_{\text{Dunkl}}} U_v + U_v \frac{T_{e_k}}{1 + \Delta_{\text{Dunkl}}} \right) \in (\mathcal{L}_{N,\infty})_0(L_2(\mathbb{R}^N, w(x)dx)).
\end{aligned}$$

Thus,

$$[R_{\text{Dunkl},j}, M_f] \in M_{\partial_j f} (1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}} + \frac{1}{2} \sum_{v \in R} v_j \kappa(v) M_{\delta_v f} (1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}} U_v -$$

$$\begin{aligned}
& -\frac{T_{e_j}}{(1 + \Delta_{\text{Dunkl}})^{\frac{1}{2}}} \cdot \sum_{k=1}^N M_{\partial_k f} \frac{T_{e_k}}{1 + \Delta_{\text{Dunkl}}} - \\
& -\frac{1}{4} \frac{T_{e_j}}{(1 + \Delta_{\text{Dunkl}})^{\frac{1}{2}}} \cdot \sum_{k=1}^N \sum_{v \in R} v_k \kappa(v) M_{\delta_v f} \left(\frac{T_{e_k}}{1 + \Delta_{\text{Dunkl}}} U_v + U_v \frac{T_{e_k}}{1 + \Delta_{\text{Dunkl}}} \right) \in \\
& \in (\mathcal{L}_{N,\infty})_0(L_2(\mathbb{R}^N, w(x)dx)).
\end{aligned}$$

The assertion follows from Lemma 9.9. $\square$

10. SPECTRAL ASYMPTOTIC FORMULA

Theorem 10.1. *Let $N > 2$. Let Π be the C^* -algebra of operators admitting principal symbol. For every $T \in \Pi \rtimes_{\alpha} G$ compactly supported from the right, we have*

$$\lim_{t \rightarrow \infty} t^{\frac{1}{N}} \mu(t, T \Delta^{-\frac{1}{2}}) = c_{\alpha} \|(\text{sym} \rtimes G)(T)\|_{L_N((L_{\infty}(\mathbb{R}^N) \otimes L_{\infty}(\mathbb{S}^{N-1})) \rtimes_{\beta})}.$$

here is argument by the chatbot, rewritten by me and checked and re-written by chatbot. so, the question is indeed reduced to matrix-valued FSZ. to be written carefully.

Definition 10.2. *Let G be a Weyl group, acting orthogonally on $\mathbb{R}^N$, and let Γ be a fixed chamber. Let $\{e_g\}_{g \in G}$ be the standard basis of $\mathbb{C}^G$. Define a surjective partial isometry $W : L_2(\mathbb{R}^N) \otimes \mathbb{C}^G \rightarrow L_2(\mathbb{R}^N)$ by setting*

$$W \left(\sum_{g \in G} \xi_g \otimes e_g \right) = \sum_{g \in G} (\xi_g|_{\Gamma}) \circ g^{-1}.$$

The crucial observation is

$$\mu(W^* T W) = \mu(T), \quad T \in B(L_2(\mathbb{R}^N)).$$

Indeed, this is true for every surjective isometry.

Lemma 10.3. *If $f \in C_c^{\infty}(\mathbb{R}^N)$, then*

$$M_f \Delta^{-\frac{1}{2}} \in \sum_{g \in G} M_{f \chi_{g\Gamma}} \Delta^{-\frac{1}{2}} M_{\chi_{g\Gamma}} + (\mathcal{L}_{N,\infty})_0(L_2(\mathbb{R}^N)).$$

Proof. This follows from Theorem 4.1 in [5]. $\square$

Lemma 10.4. *If $f \in C_c^{\infty}(\mathbb{R}^N)$ is supported away from the hyperplanes $\{\langle x, v \rangle = 0\}_{v \in R}$ and if $h \in C^{\infty}(\mathbb{S}^{N-1})$, then*

$$h\left(\frac{\nabla}{\Delta^{\frac{1}{2}}}\right) M_f \in \sum_{g \in G} M_{\chi_{g\Gamma}} h\left(\frac{\nabla}{\Delta^{\frac{1}{2}}}\right) M_{f \chi_{g\Gamma}} + \mathcal{K}(L_2(\mathbb{R}^N)).$$

Proof. Indeed, $f \chi_{g\Gamma} \in C_c^{\infty}(\mathbb{R}^N)$ for $g \in \Gamma$. Hence, **this requires a reference!!**

$$[h\left(\frac{\nabla}{\Delta^{\frac{1}{2}}}\right), M_{f \chi_{g\Gamma}}] \in \mathcal{K}(L_2(\mathbb{R}^N)), \quad g \in G.$$

The assertion follows now from the following algebraic identity.

$$h\left(\frac{\nabla}{\Delta^{\frac{1}{2}}}\right) M_f = \sum_{g \in G} h\left(\frac{\nabla}{\Delta^{\frac{1}{2}}}\right) M_{f \chi_{g\Gamma}} = \sum_{g \in G} h\left(\frac{\nabla}{\Delta^{\frac{1}{2}}}\right) M_{f \chi_{g\Gamma}} \cdot M_{\chi_{g\Gamma}} =$$

$$\begin{aligned}
&= \sum_{g \in G} M_{f\chi_{g\Gamma}} h\left(\frac{\nabla}{\Delta^{\frac{1}{2}}}\right) \cdot M_{\chi_{g\Gamma}} + \sum_{g \in G} [h\left(\frac{\nabla}{\Delta^{\frac{1}{2}}}\right), M_{f\chi_{g\Gamma}}] \cdot M_{\chi_{g\Gamma}} = \\
&= \sum_{g \in G} M_{\chi_{g\Gamma}} \cdot M_{f\chi_{g\Gamma}} h\left(\frac{\nabla}{\Delta^{\frac{1}{2}}}\right) \cdot M_{\chi_{g\Gamma}} + \sum_{g \in G} [h\left(\frac{\nabla}{\Delta^{\frac{1}{2}}}\right), M_{f\chi_{g\Gamma}}] \cdot M_{\chi_{g\Gamma}} = \\
&= \sum_{g \in G} M_{\chi_{g\Gamma}} \cdot h\left(\frac{\nabla}{\Delta^{\frac{1}{2}}}\right) M_{f\chi_{g\Gamma}} \cdot M_{\chi_{g\Gamma}} + \sum_{g \in G} M_{\chi_{g\Gamma}} \cdot [M_{f\chi_{g\Gamma}}, h\left(\frac{\nabla}{\Delta^{\frac{1}{2}}}\right)] \cdot M_{\chi_{g\Gamma}} + \\
&\quad + \sum_{g \in G} [h\left(\frac{\nabla}{\Delta^{\frac{1}{2}}}\right), M_{f\chi_{g\Gamma}}] \cdot M_{\chi_{g\Gamma}}.
\end{aligned}$$

□

Lemma 10.5. *Let $f \in C_c^\infty(\mathbb{R}^N)$ and let $g \in G$. We have*

$$\begin{aligned}
W^* M_f W &= \sum_{g \in G} M_{(f\chi_{g\Gamma}) \circ g} \otimes E_{g,g}, \\
W^* (M_{\chi_{g\Gamma}} \Delta^{-\frac{1}{2}} M_{\chi_{g\Gamma}}) W &= M_{\chi_\Gamma} \Delta^{-\frac{1}{2}} M_{\chi_\Gamma} \otimes E_{g,g}, \\
W^* M_{\chi_{g\Gamma}} h\left(\frac{\nabla}{\Delta^{\frac{1}{2}}}\right) M_{\chi_{g\Gamma}} W &= M_{\chi_\Gamma} (h \circ g) \left(\frac{\nabla}{\Delta^{\frac{1}{2}}}\right) M_{\chi_\Gamma} \otimes E_{g,g}, \\
W^* U_g W &= \sum_{g' \in G} M_{\chi_\Gamma} \otimes E_{gg',g'}.
\end{aligned}$$

Proof. The third assertion follows from the equality

$$U_g^{-1} h\left(\frac{\nabla}{\Delta^{\frac{1}{2}}}\right) U_g = (h \circ g) \left(\frac{\nabla}{\Delta^{\frac{1}{2}}}\right).$$

The other assertions are obvious. □

Lemma 10.6. *If $f \in C_c^\infty(\mathbb{R}^N)$ is supported away from the hyperplanes $\{\langle x, v \rangle = 0\}_{v \in R}$ and if $h \in C^\infty(\mathbb{S}^{N-1})$, then*

$$\begin{aligned}
&W^* h\left(\frac{\nabla}{\Delta^{\frac{1}{2}}}\right) M_f \Delta^{-\frac{1}{2}} W \in \\
&\in \sum_{g \in G} \left[M_{\chi_\Gamma} (h \circ g) \left(\frac{\nabla}{\Delta^{\frac{1}{2}}}\right) M_{(f\chi_{g\Gamma}) \circ g} \Delta^{-\frac{1}{2}} M_{\chi_\Gamma} \right] \otimes E_{g,g} + (\mathcal{L}_{N,\infty})_0 (L_2(\mathbb{R}^N) \otimes \mathbb{C}^G).
\end{aligned}$$

Proof. We write $f = f_1 f_2$, where $f_1, f_2 \in C_c^\infty(\mathbb{R}^N)$ are supported away from the hyperplanes $\{\langle x, v \rangle = 0\}_{v \in R}$. The assertion follows from Lemma 10.3 (applied with f_2 in place of f), Lemma 10.4 (applied with f_1 in place of f) and Lemma 10.5. □

Lemma 10.7. *If $f \in C_c^\infty(\mathbb{R}^N)$ is supported away from the hyperplanes $\{\langle x, v \rangle = 0\}_{v \in R}$ and if $h \in C^\infty(\mathbb{S}^{N-1})$, then*

$$\begin{aligned}
&W^* h\left(\frac{\nabla}{\Delta^{\frac{1}{2}}}\right) M_f \Delta^{-\frac{1}{2}} U_g W \in \\
&\in \sum_{g' \in G} \left[(h \circ g') \left(\frac{\nabla}{\Delta^{\frac{1}{2}}}\right) M_{(f\chi_{g'\Gamma}) \circ g'} \Delta^{-\frac{1}{2}} \right] \otimes E_{g',g^{-1}g'} + (\mathcal{L}_{N,\infty})_0 (L_2(\mathbb{R}^N) \otimes \mathbb{C}^G).
\end{aligned}$$

Proof. The assertion follows from Lemma 10.6 and the facts (see the proof of Lemma 10.4 for the first one and Theorem 4.1 in [5] for the second one) that

$$M_{\chi_\Gamma}(h \circ g)\left(\frac{\nabla}{\Delta^{\frac{1}{2}}}\right)M_F - (h \circ g)\left(\frac{\nabla}{\Delta^{\frac{1}{2}}}\right)M_F \in \mathcal{K}(L_2(\mathbb{R}^N)),$$

$$M_F \Delta^{-\frac{1}{2}} M_{\chi_\Gamma} - M_{F\chi_\Gamma} \Delta^{-\frac{1}{2}} \in (\mathcal{L}_{N,\infty})_0(L_2(\mathbb{R}^N)),$$

for every $F \in C_c^\infty(\mathbb{R}^N)$ is supported away from the hyperplanes $\{\langle x, v \rangle = 0\}_{v \in R}$. $\square$

Theorem 10.8. *Let $N > 2$. Let Π be the C^* -algebra of operators admitting principal symbol. For every $S \in \Pi \otimes M_m(\mathbb{C})$, compactly supported from the right, we have*

$$\lim_{t \rightarrow \infty} t^{\frac{1}{N}} \mu(t, S(\Delta^{-\frac{1}{2}} \otimes 1)) = c_N \|(\text{sym} \otimes \text{id})(S)\|_{L_N((L_\infty(\mathbb{R}^N) \bar{\otimes} L_\infty(\mathbb{S}^{N-1})) \otimes M_m(\mathbb{C}))}.$$

Proof of Theorem 10.1. Suppose first

$$T = \sum_{g \in G} \sum_{k=1}^K h_{k,g} \left(\frac{\nabla}{\Delta^{\frac{1}{2}}} \right) M_{f_{k,g}} U_g$$

with $h_{k,g} \in C^\infty(\mathbb{S}^{N-1})$ and with $f_{k,g} \in C_c^\infty(\mathbb{R}^N)$ supported away from the hyperplanes $\{\langle x, v \rangle = 0\}_{v \in R}$. Using Lemma 10.7, we write

$$W^*TW \in S(\Delta^{-\frac{1}{2}} \otimes 1) + (\mathcal{L}_{N,\infty})_0(L_2(\mathbb{R}^N) \otimes \mathbb{C}^G),$$

where

$$S = \sum_{g,g' \in G} \sum_{k=1}^K \left[(h_{k,g} \circ g') \left(\frac{\nabla}{\Delta^{\frac{1}{2}}} \right) M_{(f_{k,g}\chi_{g'\Gamma}) \circ g'} \right] \otimes E_{g',g^{-1}g'}.$$

Obviously, $S \in \Pi \otimes M_G(\mathbb{C})$ is compactly supported from the right. We have

$$(\text{sym} \otimes \text{id})(S) = \sum_{g,g' \in G} \sum_{k=1}^K \left[(h_{k,g} \circ g') \otimes ((f_{k,g}\chi_{g'\Gamma}) \circ g') \right] \otimes E_{g',g^{-1}g'}.$$

$\square$

APPENDIX A. ZHIJIE LEMMA

Lemma A.1. *Let $1 < s < \infty$, and let $\mathbb{D}^0$ be the standard dyadic grid in $\mathbb{R}^N$. Suppose that $F \in L^s(\mathbb{R}^N)$. Then there exists a constant $C > 0$ such that*

$$\left\| \left\{ |Q|^{-1/s'} \int_Q |F(x)| dx \right\}_{Q \in \mathbb{D}^0} \right\|_{\ell^{s,\infty}} \leq C \|F\|_{L^s(\mathbb{R}^N)}.$$

Proof. Let $\mathcal{E} \subset \mathbb{D}^0$ be finite and set

$$G_{\mathcal{E}}(x) := \sum_{Q \in \mathcal{E}} |Q|^{-1/s'} \chi_Q(x).$$

Since the dyadic cubes containing any fixed x are totally ordered by inclusion, for each $x \in \bigcup_{Q \in \mathcal{E}} Q$ we may choose a cube $Q_x \in \mathcal{E}$ of minimal volume such that

$x \in Q_x$. Every $Q \in \mathcal{E}$ containing x is then a dyadic ancestor of Q_x . Since the volume increases by the factor 2^N at each generation,

$$G_{\mathcal{E}}(x) = \sum_{\substack{Q \in \mathcal{E} \\ x \in Q}} |Q|^{-1/s'} \leq |Q_x|^{-1/s'} \sum_{k=0}^{\infty} 2^{-kN/s'} \lesssim_{N,s} \sup_{Q \in \mathcal{E}} |Q|^{-1/s'} \chi_Q(x).$$

Consequently,

$$\|G_{\mathcal{E}}\|_{L^{s'}(\mathbb{R}^N)}^{s'} \lesssim_{N,s} \int_{\mathbb{R}^N} \sup_{Q \in \mathcal{E}} |Q|^{-1} \chi_Q(x) dx \leq \int_{\mathbb{R}^N} \sum_{Q \in \mathcal{E}} |Q|^{-1} \chi_Q(x) dx = \#\mathcal{E}.$$

and hence,

$$\|G_{\mathcal{E}}\|_{L^{s'}(\mathbb{R}^N)} \lesssim_{N,s} (\#\mathcal{E})^{1/s'}.$$

By Hölder's inequality,

$$\sum_{Q \in \mathcal{E}} |Q|^{-1/s'} \int_Q |F(x)| dx = \int_{\mathbb{R}^N} |F(x)| G_{\mathcal{E}}(x) dx \lesssim_{N,s} \|F\|_{L^s(\mathbb{R}^N)} (\#\mathcal{E})^{1/s'}.$$

Fix $t > 0$ and let $\mathcal{E}$ be any finite subcollection of the dyadic cubes satisfying

$$|Q|^{-1/s'} \int_Q |F(x)| dx > t.$$

Then

$$t \#\mathcal{E} < \sum_{Q \in \mathcal{E}} |Q|^{-1/s'} \int_Q |F(x)| dx \lesssim_{N,s} \|F\|_{L^s(\mathbb{R}^N)} (\#\mathcal{E})^{1/s'},$$

so

$$\#\mathcal{E} \lesssim_{N,s} t^{-s} \|F\|_{L^s(\mathbb{R}^N)}^s.$$

Since $\mathcal{E}$ is arbitrary,

$$\#\left\{Q \in \mathbb{D}^0 : |Q|^{-1/s'} \int_Q |F(x)| dx > t\right\} \lesssim_{N,s} t^{-s} \|F\|_{L^s(\mathbb{R}^N)}^s, \quad t > 0.$$

The asserted weak ℓ^s estimate follows. This completes the proof of Lemma A.1. $\square$

REFERENCES

- [1] Nicolas Bourbaki. *Lie groups and Lie algebras. Chapters 4–6*. Elements of Mathematics (Berlin). Springer-Verlag, Berlin, 2002. Translated from the 1968 French original by Andrew Pressley. [3](#)
- [2] Jacek Dziubański and Agnieszka Hejna. Remark on atomic decompositions for the Hardy space H^1 in the rational Dunkl setting. *Studia Math.*, 251(1):89–110, 2020. [10](#)
- [3] Z. Fan, M. Lacey, and J. Li. Schatten classes and commutators of Riesz transform on Heisenberg group and applications. *J. Fourier Anal. Appl.*, 29(2):Paper No. 17, 28, 2023. [3](#)
- [4] Z. Fan, M. Lacey, J. Li, M.N. Vempati, and B.D. Wick. Besov space, Schatten classes, and commutators of Riesz transforms associated with the Neumann Laplacian. *Michigan Math. J.*, 75(1):199–224, 2025. [3](#)
- [5] R.L. Frank, F. Sukochev, and D. Zanin. Asymptotics of singular values for quantum derivatives. *Trans. Amer. Math. Soc.*, 376(3):2047–2088, 2023. [30](#), [32](#)
- [6] Y. Han, M. Lee, J. Li, and B.D. Wick. Riesz transforms and commutators in the Dunkl setting. *Anal. Math. Phys.*, 14(3):Paper No. 46, 32, 2024. [14](#)
- [7] S. Helgason. *Differential geometry, Lie groups, and symmetric spaces*, volume 80 of *Pure and Applied Mathematics*. Academic Press, Inc. [Harcourt Brace Jovanovich, Publishers], New York-London, 1978. [2](#)
- [8] T. Hytonen. A_∞ -invariance of oscillatory norms and schatten characterisations of commutators. [18](#)

- [9] T. Hytonen. Schatten properties of commutators on metric spaces. Preprint available at <https://arxiv.org/abs/2411.02613>, 2024. [3](#), [4](#), [18](#)
- [10] M.T. Lacey, J. Li, and B.D. Wick. Schatten classes and commutator in the two weight setting, I. Hilbert transform. *Potential Anal.*, 60(2):875–894, 2024. [3](#)
- [11] J. Li, X. Xiong, and F. Yang. Schatten properties of Calderón–Zygmund singular integral commutator on stratified Lie groups. *J. Math. Pures Appl. (9)*, 188:73–113, 2024. [3](#)
- [12] R. Rochberg and S. Semmes. Nearly weakly orthonormal sequences, singular value estimates, and Calderon-Zygmund operators. *J. Funct. Anal.*, 86(2):237–306, 1989. [3](#)
- [13] Z. Wei and H. Zhang. Complex median method and Schatten class membership of commutators. Preprint available at <https://arxiv.org/abs/2411.05810>, 2024. [3](#)
- [14] Z. Wei and H. Zhang. On the boundedness and Schatten class property of noncommutative martingale paraproducts and operator-valued commutators. Preprint available at <https://arxiv.org/abs/2407.14988>, 2024. [3](#)

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